

Proudly supported by



edunet
foundation



LAB MANUAL

Explore SAP Analytics Cloud by Various Features.

6.1 Practical- Getting Started Create a Standalone Dataset

Objective:

The objective of this practical is to gain hands-on experience with creating and managing data models and dimensions within **SAP Analytics Cloud (SAC)**. Learners will begin by creating a **standalone dataset**, followed by the creation of a custom **Stores dimension** including its **hierarchy and properties** to better organize and structure data. The next step involves **importing and preparing data** for a more advanced dimension, such as the **Super Stores dimension**, ensuring data consistency and relevance. Participants will also learn how to **create an Import Analytic Model**, a more structured and reusable model for deeper analysis. As part of the model-building process, they will **import and prepare data within the analytic model**, ensuring correct mapping of dimensions and measures. Finally, learners will be guided to **create calculated measures** inside the model, enabling dynamic and custom business metrics such as profit margins or growth percentages. By the end of this practical, users will have a strong foundational understanding of building datasets, dimensions, and analytic models in SAC, which are critical for effective data analysis and story creation.

Equipment Required:

1. Laptop/Desktop Computer with Processor: Intel i3 or above, RAM: 4 GB (8 GB recommended), Internet connectivity, Screen resolution: 1366x768 or higher
2. Stable High-Speed Internet Connection Required for accessing the SAP Analytics Cloud platform (as it's cloud-based). Minimum speed: 4 Mbps (per device)

Prerequisites:

1. Basic Knowledge of SAP Analytics Cloud (SAC), Familiarity with the SAC user interface. Understanding of terms like datasets, models, measures, and dimensions.
2. Access to a working SAP Analytics Cloud tenant (enterprise or trial), User role must allow creation of datasets, models, and dimensions.
3. Sample Data Files Ready: Excel or CSV files containing data for, Store information (e.g., Store ID, Location, Region) Super Store sales or transaction data.
4. Basic understanding of data types (e.g., string, date, number), Familiarity with concepts like hierarchies, dimensions, and calculated fields

Problem Statement:

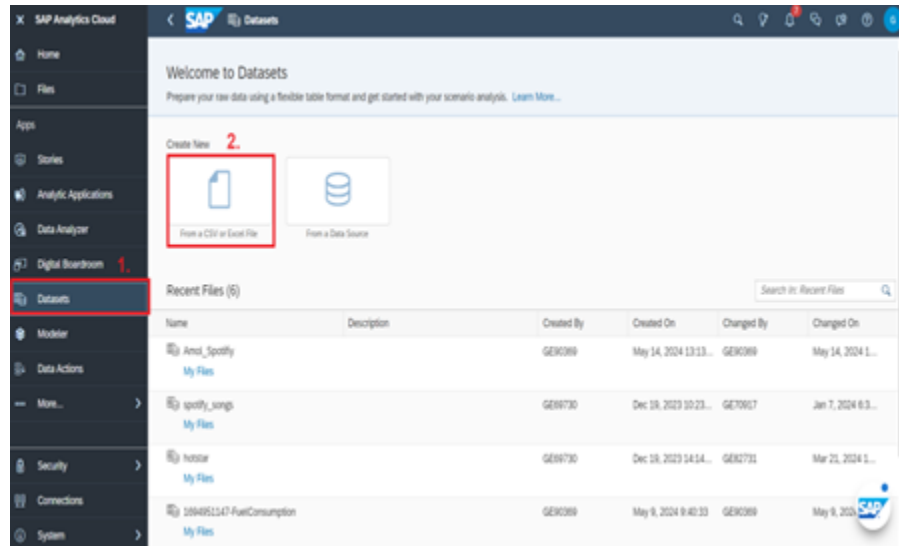
Getting Started – Creating Dataset, Dimensions, Analytic Models, and Calculated Measures in SAP Analytics Cloud.

Procedure:

Disclaimer: The content is curated from online/offline resources and used for educational purpose only

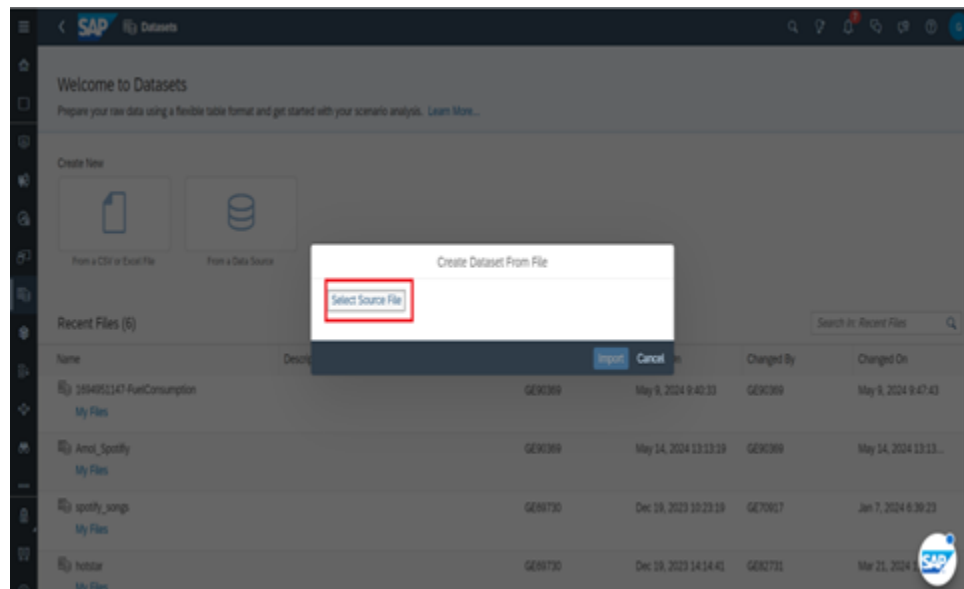
Step-1: Directory to create datasets

1. Select **Datasets** option
2. Click on **From csv or excel file**



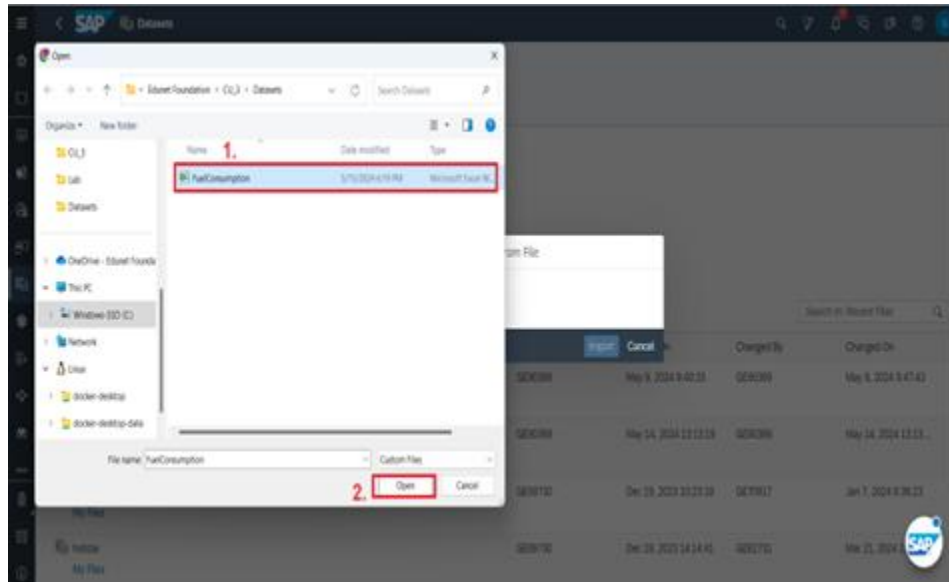
Step-1: Directory to create datasets

1. Select the Source file



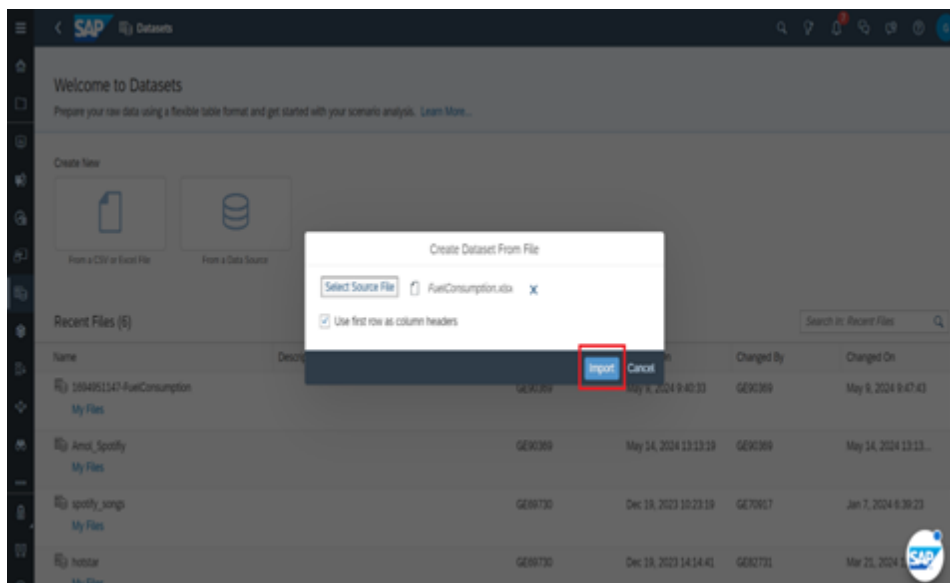
Step-1: Directory to create datasets

1. Select **Fuel Consumption Datasets** option
2. Click on **Open**



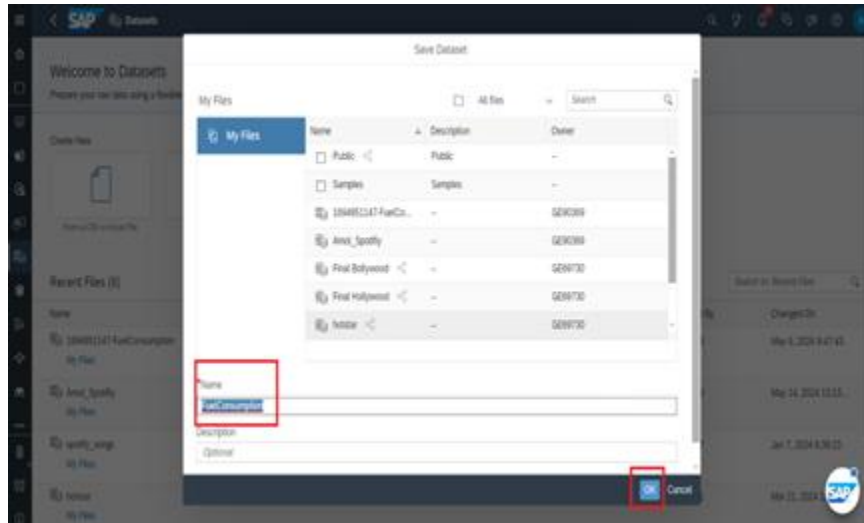
Step-1: Directory to create datasets

1. Select **Import** option



Step-1: Directory to create datasets

1. Enter the **Name** as **FuelConsumption**
2. Click on **OK**



Step-2: Handle the missing values

Your dataset will be load in few minutes.

Wonderful, see this is the beauty of SAC were

1. Automatically detect the datatype of the columns
2. Detect the 4 warnings
3. To solve these warnings, click on **Issues**

	Year	MAKE-M	VEHICLE	ENGINE	CYLINDERS	TRANSM	FUEL	FUEL CO.
1	2000	ACURA-1.6L	COMPACT	1.6	4	A4	X	10.5
2	2000	ACURA-1.6L	COMPACT	1.6	4	M5	X	9.80000000X
3	2000	ACURA-3.2TL	MID-SIZE	3.2	6	A55	Z	13.7
4	2000	ACURA-3.5RL	MID-SIZE	3.5	6	A4	Z	15
5	2000	ACURA-INTegra	SUBCOMPACT	1.8	4	A4	X	11.4
6	2000	ACURA-INTegra	SUBCOMPACT	1.8	4	M5	X	10.6
7	2000	ACURA-INTegra	SUBCOMPACT	1.8	4	M5	Z	10.8
8	2000	ACURA-NSX	SUBCOMPACT	3	6	A54	Z	15.2
9	2000	ACURA-NSX	SUBCOMPACT	3.2	6	M6	Z	15.4
10	2000	AUDI-A4	COMPACT	1.8	4	A5	NA	12.9
11	2000	AUDI-A4	COMPACT	1.8	4	M5	Z	11.1
12	2000	AUDI-A4	COMPACT	2.8	6	A5	Z	14.6
13	2000	AUDI-A4	COMPACT	2.8	6	M5	Z	13.2
14	2000	AUDI-A4 QUATT	COMPACT	1.8	4	A5	Z	13.6
15	2000	AUDI-A4 QUATT	COMPACT	1.8	4	M5	Z	12.1
16	2000	AUDI-A4 QUATT	COMPACT	2.8	6	A5	Z	14.9

Step-2: Handle the missing values

See getting the issue because of the NA value present in dataset. Let's try to fix it.

1. Select **NA** option
2. Click on **Replace**
3. Enter the **"2000"** value
4. Click on **tick** or press **Enter**.

You will see you have successfully transformed the values.

5. Click on the **Dataset Overview**.

Repeat the same steps to replace the values in remaining columns

	Year	MAKE-M	VEHICLE	ENGINE	CYLINDERS	TRANSM	FUEL	FUEL CO.
421	2000	DES-BENG	MINICOMPACT	4.3	8	A5	Z	15.2
1	2000	ACURA-1.6L	COMPACT	1.6	4	A4	X	10.5
2	2000	ACURA-1.6L	COMPACT	1.6	4	M5	X	9.80000000X
3	2000	ACURA-3.2TL	MID-SIZE	3.2	6	A55	Z	13.7
4	2000	ACURA-3.5RL	MID-SIZE	3.5	6	A4	Z	15
5	2000	ACURA-INTegra	SUBCOMPACT	1.8	4	A4	X	11.4
6	2000	ACURA-INTegra	SUBCOMPACT	1.8	4	M5	X	10.6
7	2000	ACURA-INTegra	SUBCOMPACT	1.8	4	M5	Z	10.8
8	2000	ACURA-NSX	SUBCOMPACT	3	6	A54	Z	15.2
9	2000	ACURA-NSX	SUBCOMPACT	3.2	6	M6	Z	15.4
10	2000	AUDI-A4	COMPACT	1.8	4	A5	NA	12.9
11	2000	AUDI-A4	COMPACT	1.8	4	M5	Z	11.1
12	2000	AUDI-A4	COMPACT	2.8	6	A5	Z	14.6
13	2000	AUDI-A4	COMPACT	2.8	6	M5	Z	13.2
14	2000	AUDI-A4 QUATT	COMPACT	1.8	4	A5	Z	13.6
15	2000	AUDI-A4 QUATT	COMPACT	1.8	4	M5	Z	12.1

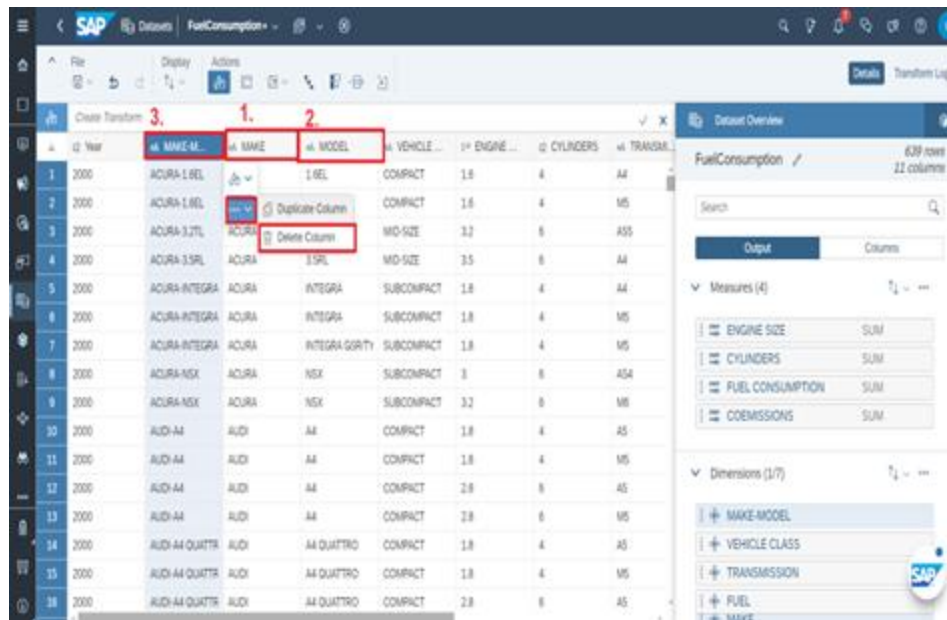
Step-3: Split the columns

1. See all issues resolved
2. Click on **MAKE-MODEL** column
3. Click on the **transform** icon
4. Click on the **Split on "-"** option

See the columns splits into new columns. Now let's edit the names of columns and delete the old one.

Step-4: Edit the name of the columns

1. Double click on the new column and rename as **MAKE**
2. Double click on the new column and rename as **MODEL**
3. Delete the **MAKE-MODEL** after clicking on Same column you will get **3 dots** and **Delete** the column

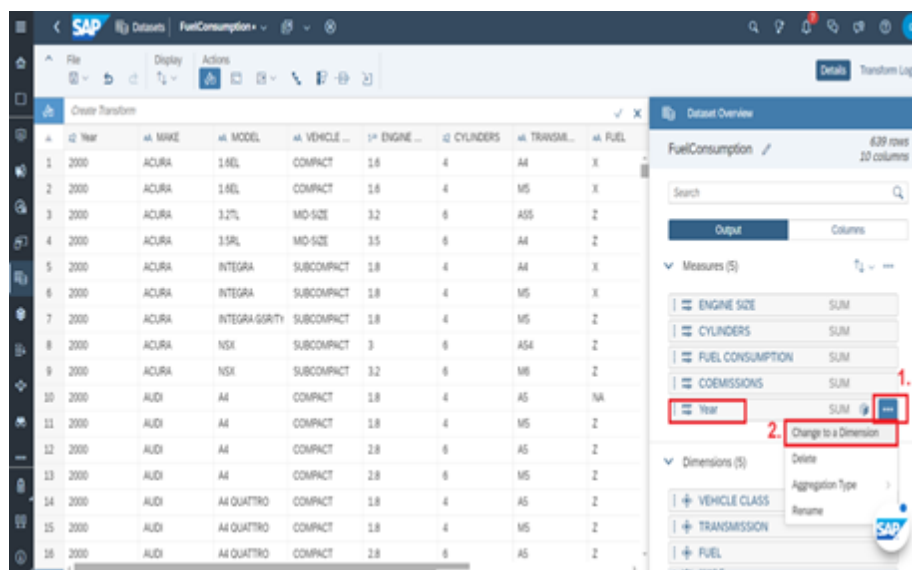


Step-5: Change to Dimension

As the Date column in the Measures

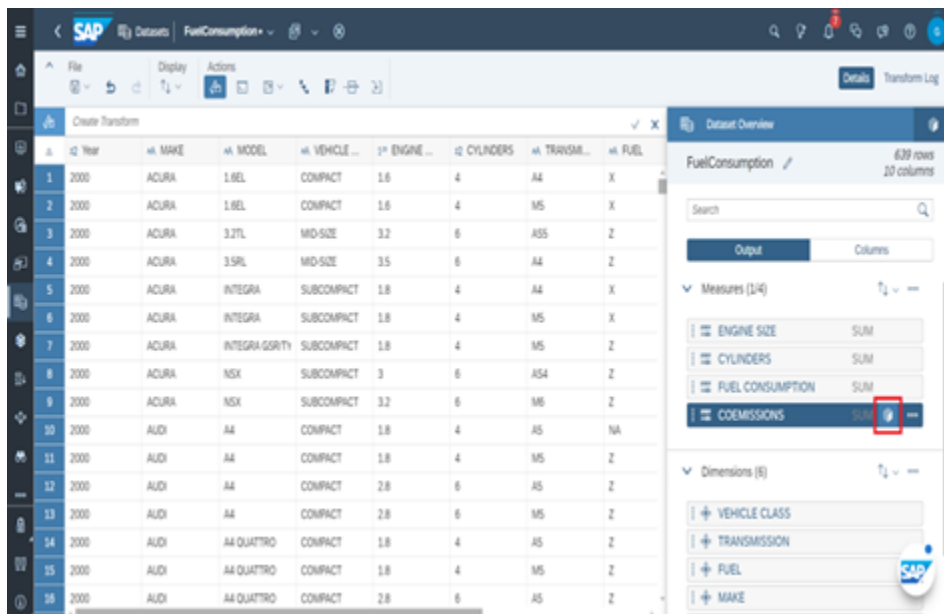
It should be in the Dimension.

1. Select **3 dots (Move cursor)** inline in Year option under the Measures
2. Click on **Change to a Dimension**
3. Scroll down, you will see the **Date** option availed under the Dimension



Step-6: Check the distribution of data

1. Click on the  option in line with the COEMISSIONS



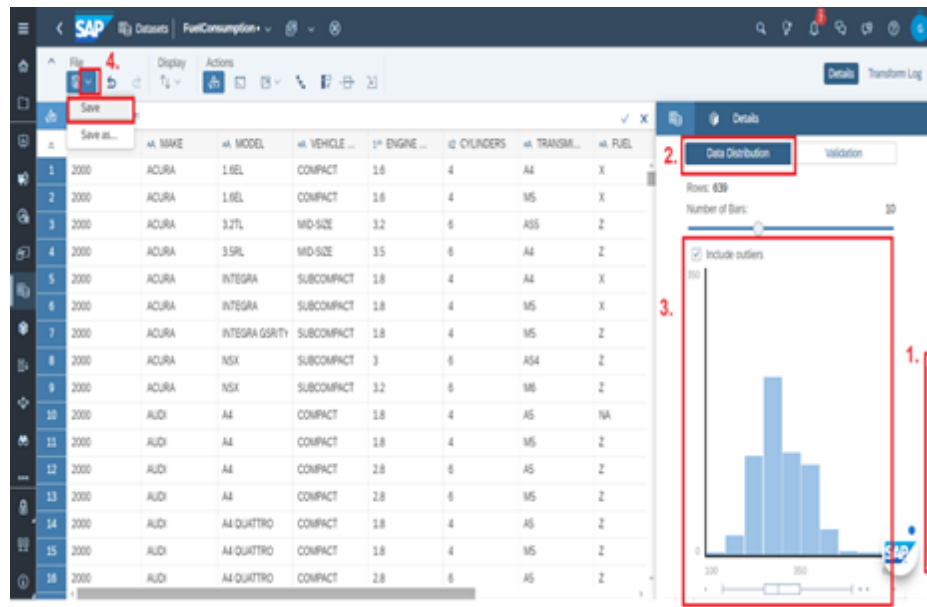
Year	MAKE	MODEL	VEHICLE ...	1 st ENGINE ...	2 nd CYLINDERS	TRANSM...	FUEL
2000	ACURA	1.8EL	COMPACT	1.8	4	A4	X
2000	ACURA	1.8EL	COMPACT	1.8	4	M5	X
2000	ACURA	3.2TL	MID-SIZE	3.2	6	A35	Z
2000	ACURA	3.5RL	MID-SIZE	3.5	6	A4	Z
2000	ACURA	INTEGRA	SUBCOMPACT	1.8	4	A4	X
2000	ACURA	INTEGRA	SUBCOMPACT	1.8	4	M5	X
2000	ACURA	INTEGRA GSRTH	SUBCOMPACT	1.8	4	M5	Z
2000	ACURA	NSX	SUBCOMPACT	3	6	A34	Z
2000	ACURA	NSX	SUBCOMPACT	3.2	6	M6	Z
2000	AUDI	A4	COMPACT	1.8	4	A5	NA
2000	AUDI	A4	COMPACT	1.8	4	M5	Z
2000	AUDI	A4	COMPACT	2.8	6	A5	Z
2000	AUDI	A4	COMPACT	2.8	6	M5	Z
2000	AUDI	A4 QUATTRO	COMPACT	1.8	4	A5	Z
2000	AUDI	A4 QUATTRO	COMPACT	1.8	4	M5	Z
2000	AUDI	A4 QUATTRO	COMPACT	2.8	6	A5	Z

Step-6: Check the distribution of data

1. Scroll Down
2. Click on the **Data Distribution**
3. See the **Histogram and Box Plot**. You can modify the number of bars.
4. Click on **down arrow** and Save the Dataset.

In this way you can create dataset with the excel files.

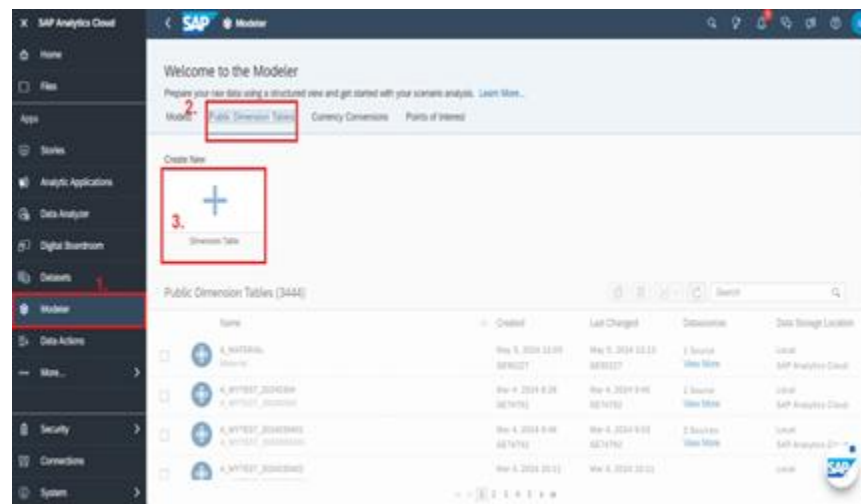
Now let's understand the dimensions in details



6.2 Practical - Create the Stores Dimension, Hierarchy, and Properties

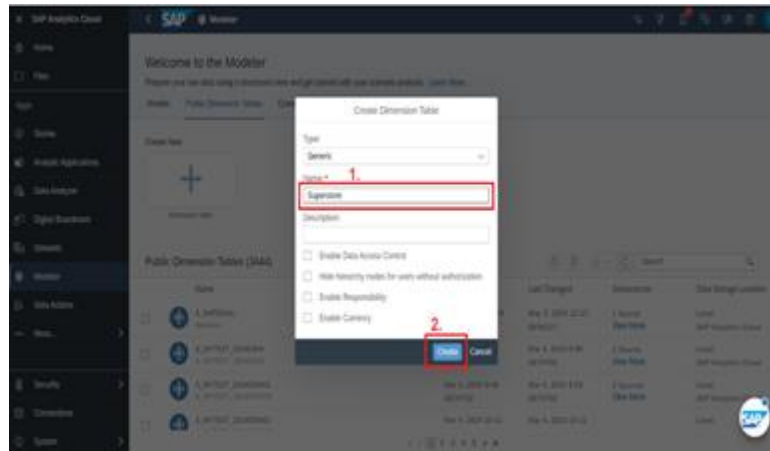
Step-1: Create the public dimension table

1. Go to Home tab, select **Modular** Option
2. Select **Public Dimension Table** option
3. Click on the **+** option



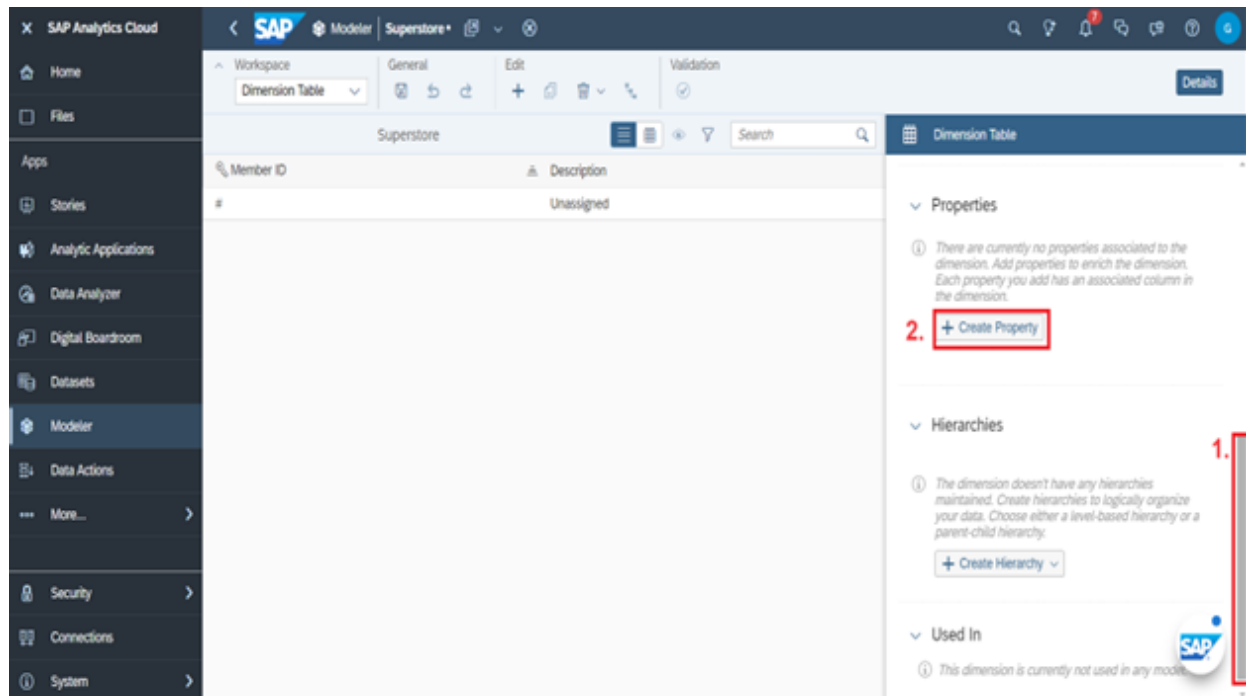
Step-1: Create the public dimension table

1. Enter the Name as **Superstore**
2. Click on **Create**



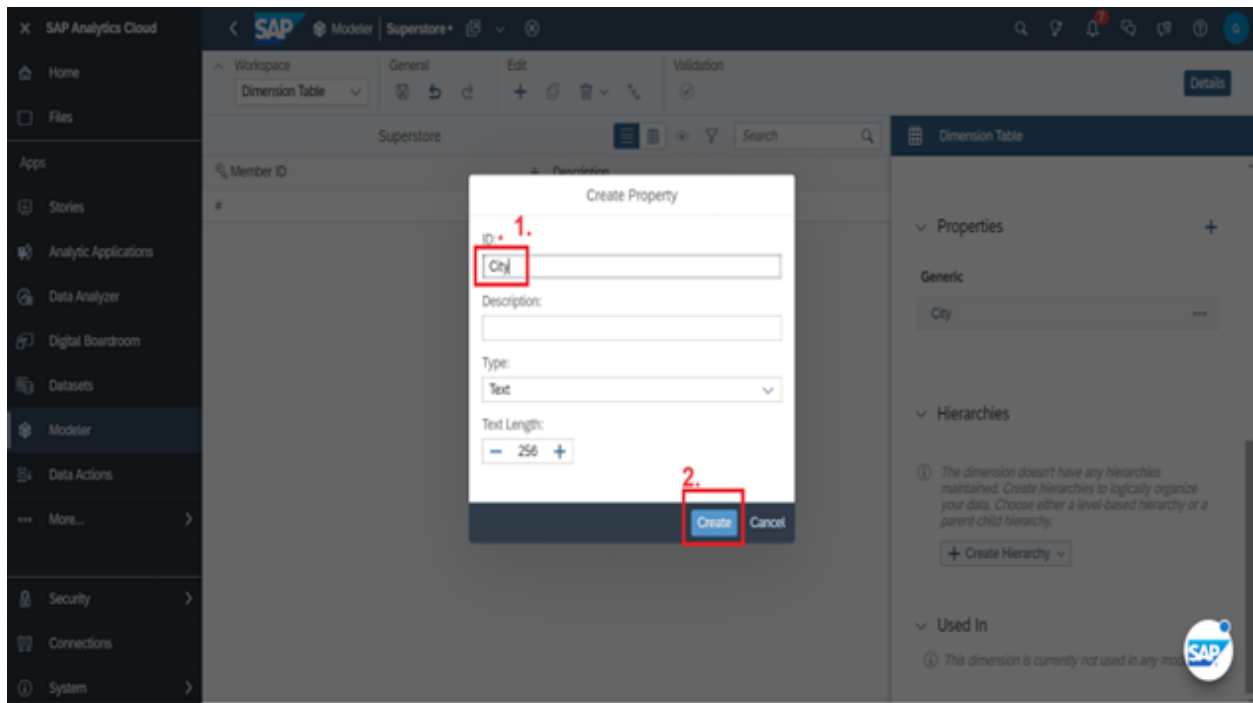
Step-2: Create Properties

1. Scroll down
2. Click on **Create Property**



Step-2: Create Properties

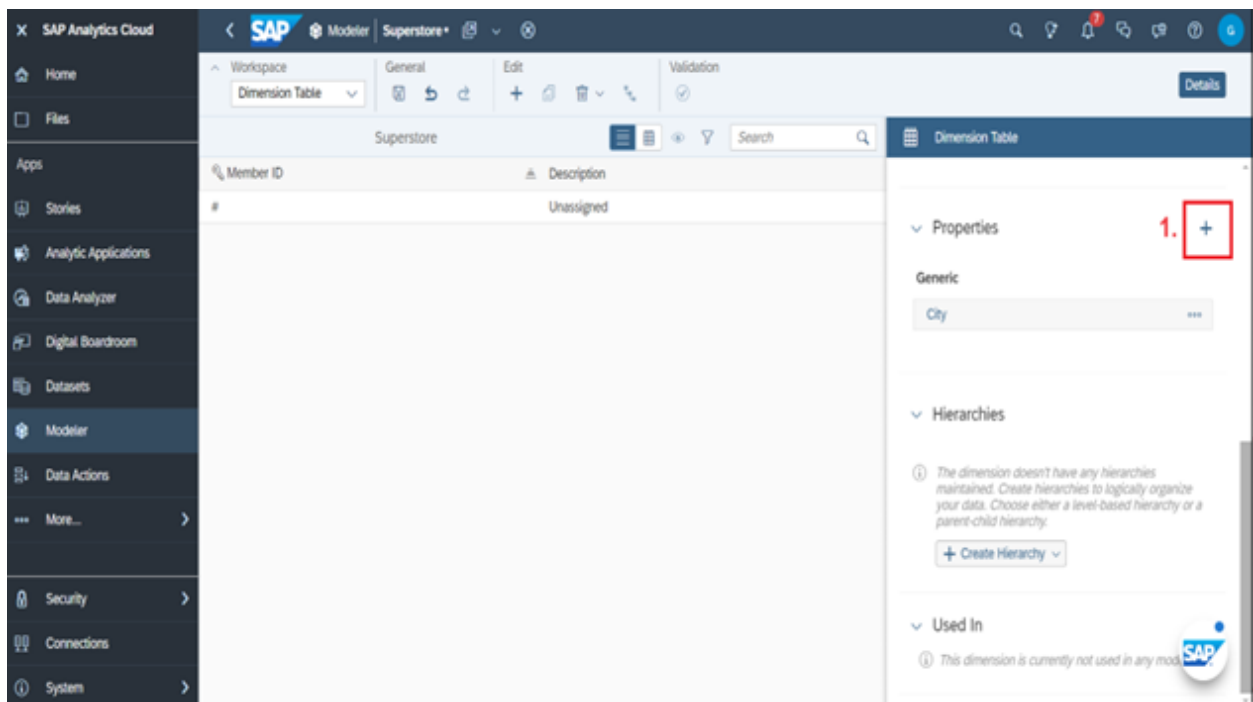
1. Enter the ID as **City**
2. Click on **Create**



Step-2: Create Properties

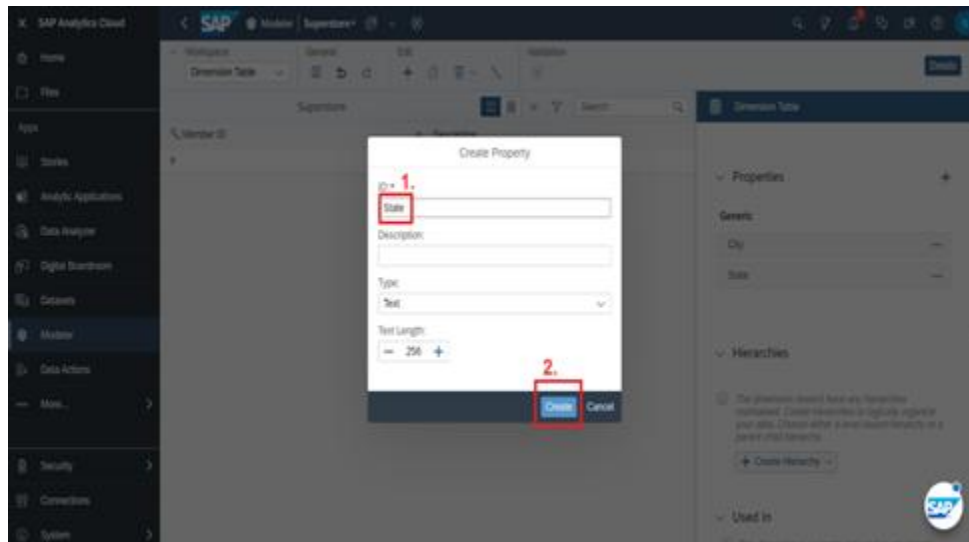
1. Click on the + button

Again, to more properties



Step-2: Create Properties

1. Enter the ID as **State**
2. Click on the **Create**

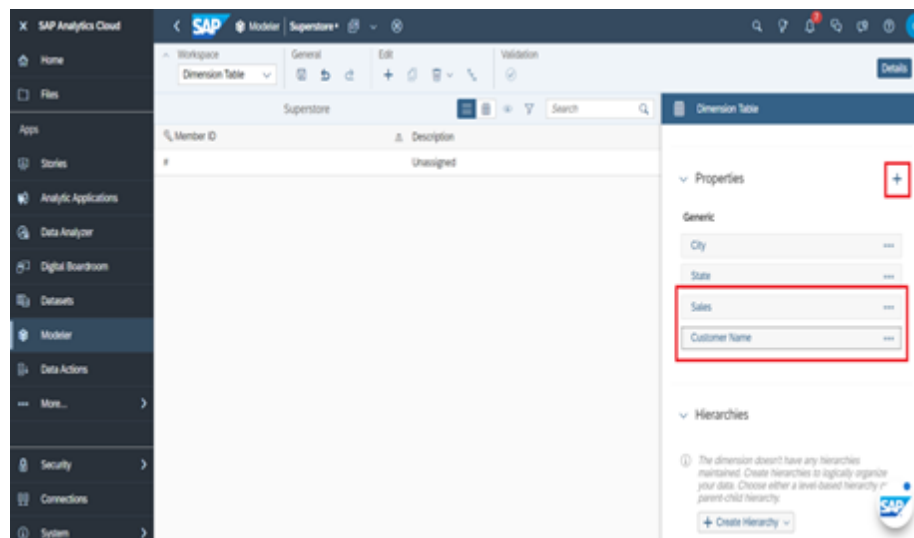


Step-2: Create Properties

1. Click on the + button
2. Add **Sales** and the

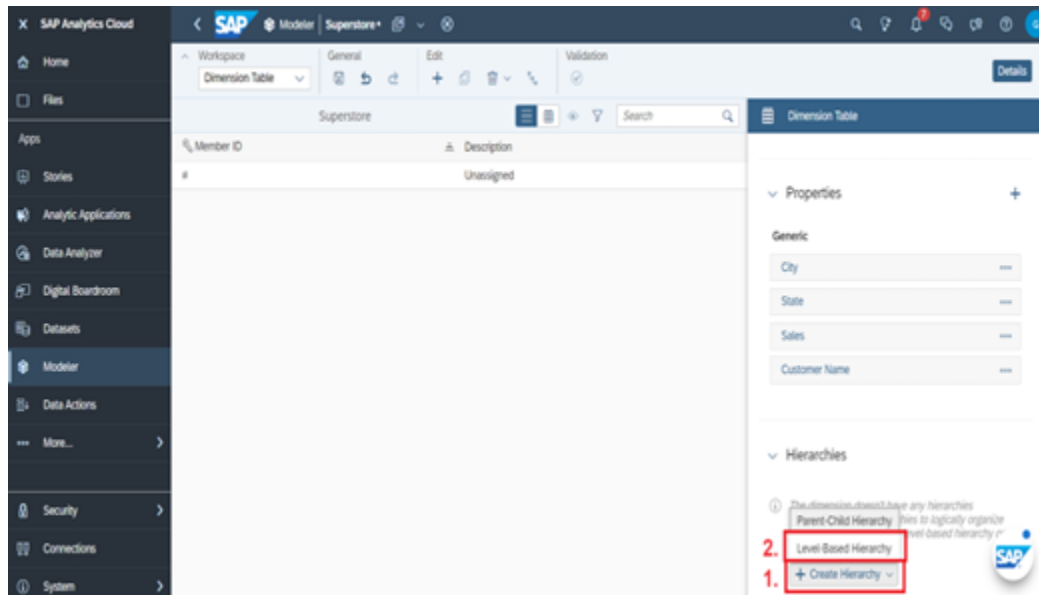
Customer Name property in

Similar manner



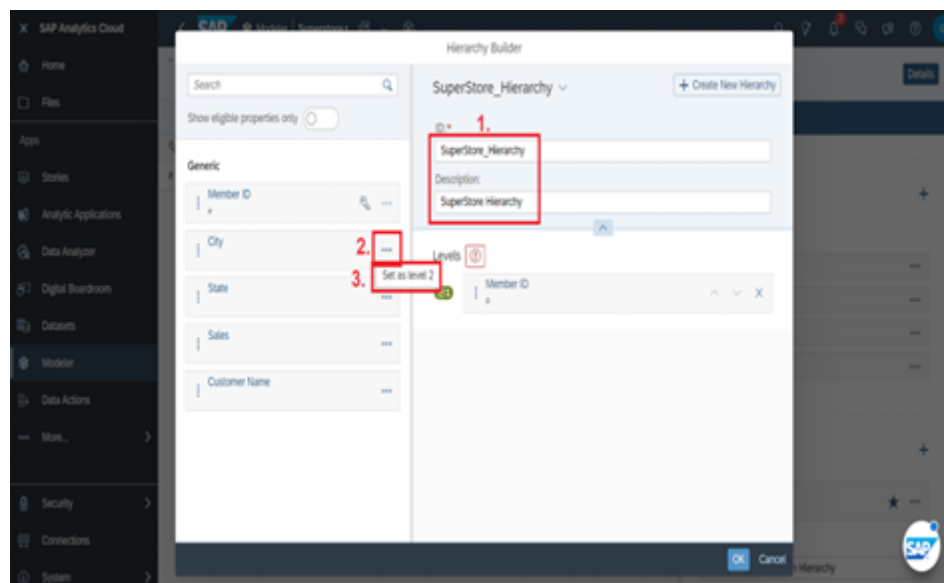
Step-3: Create Hierarchy

1. Click on the **Create Hierarchy** button
2. Click on the **Level-Based Hierarchy**



Step-3: Create Hierarchy

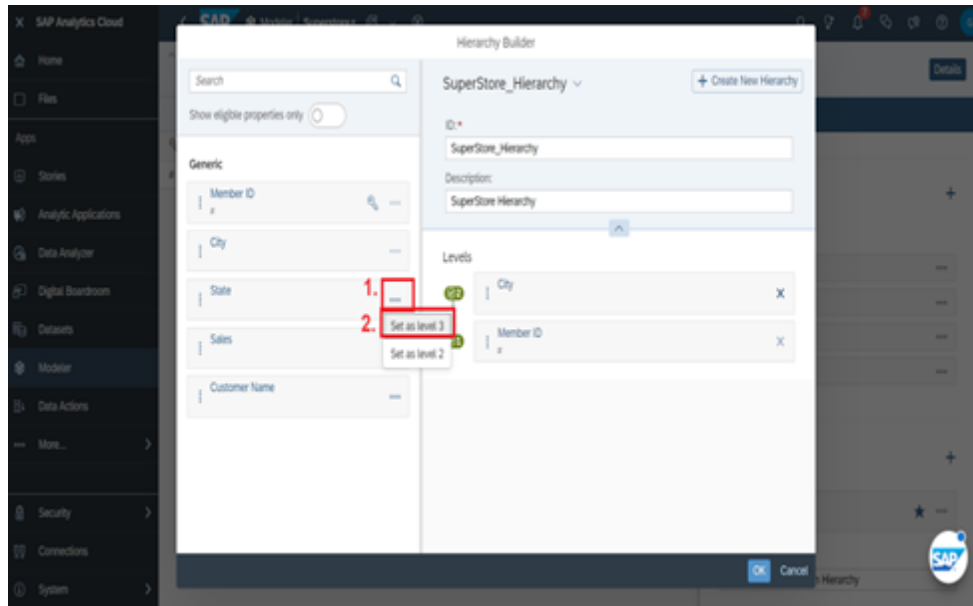
1. Enter the ID as **SuperStore_Hierarchy** same in the description as well
2. Click on the **3 dots**
3. Click on the **Set as Level 2**



Step-3: Create Hierarchy

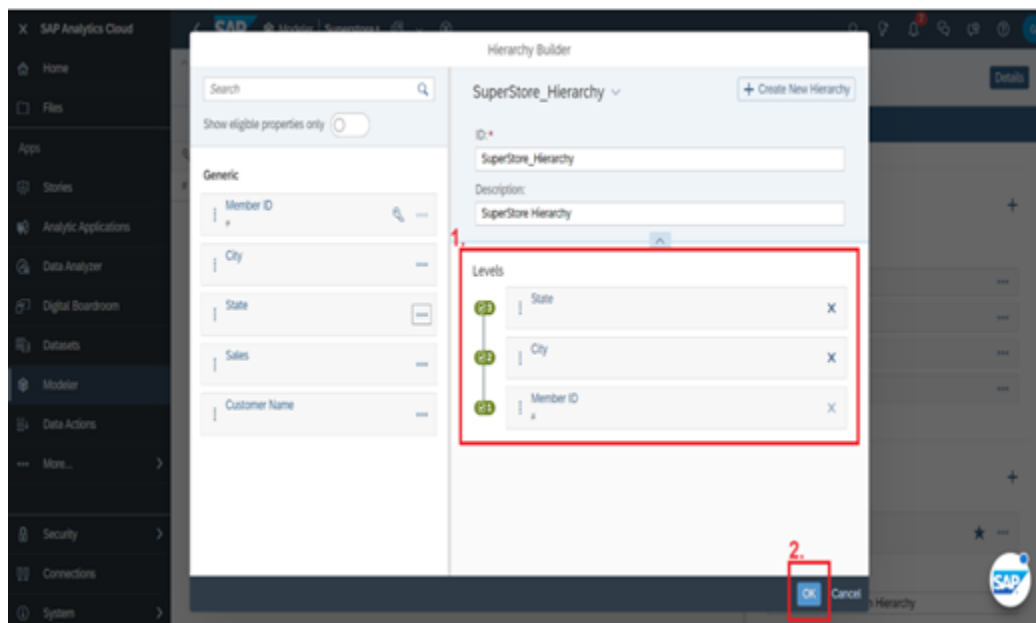
Disclaimer: The content is curated from online/offline resources and used for educational purpose only

1. Click on the **3 dots** in line with the State
2. Click on the **Set as Level3**



Step-3: Create Hierarchy

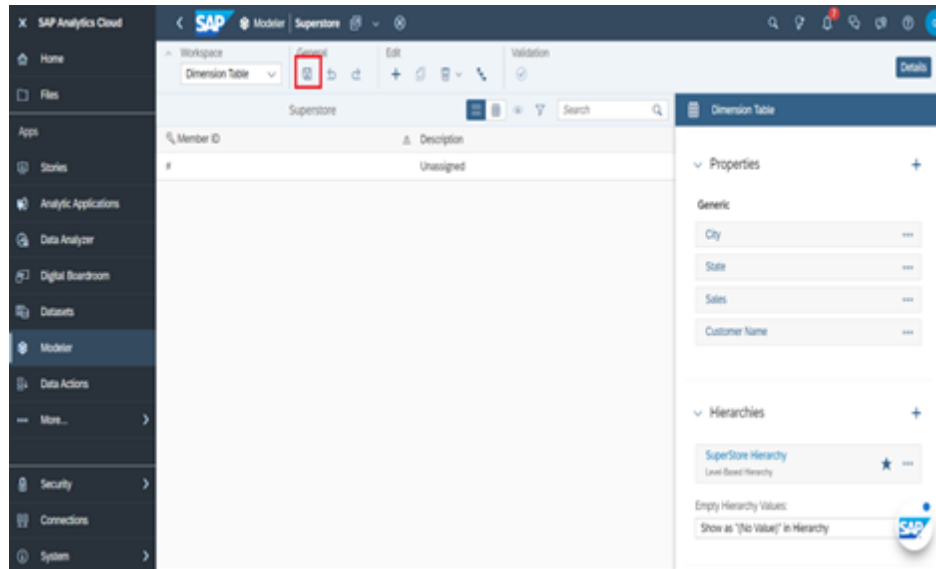
1. **Congratulations, see you have successfully created the Hierarchy in increasing order of granularity.**
2. Click on **OK** button



Step-4: Save file

1. Click on the **Save** option

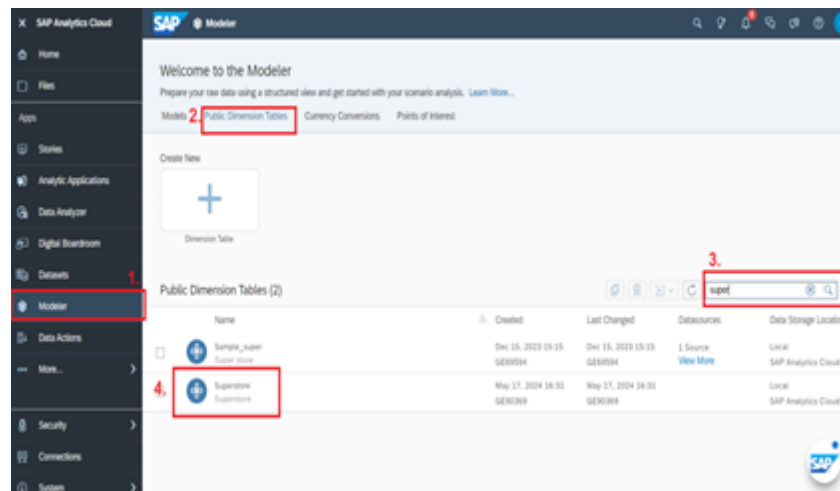
You have successfully created the superstore dimensions, properties and the hierarchy



6.3 Practical - Import and Prepare Data for the Super Stores Dimension

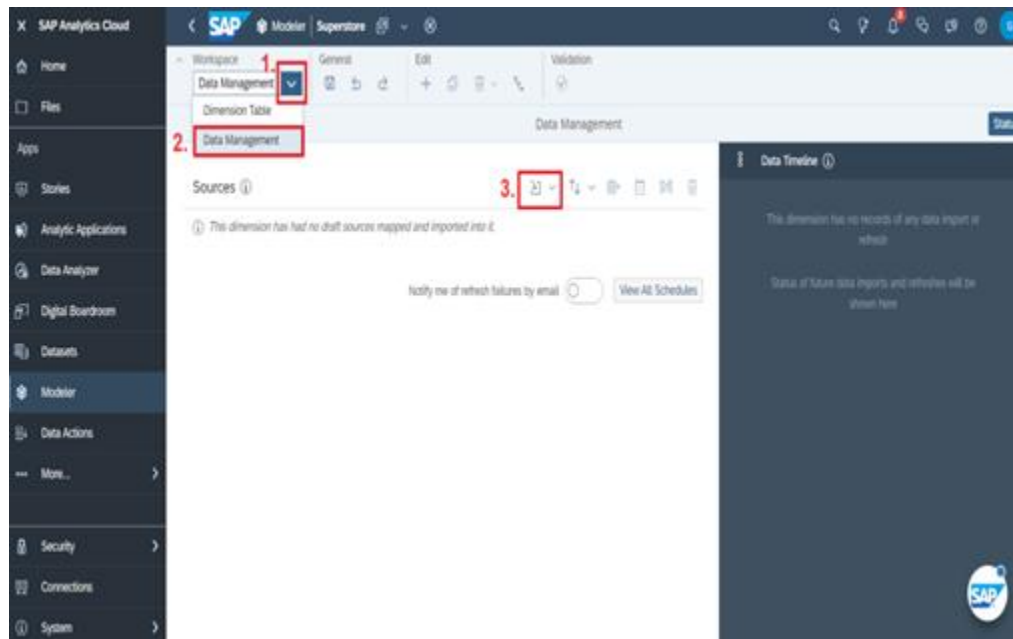
Step-1: Navigate to the Super Store dimension table

1. Select the **Modular**
2. Click on the **Public Dimension Table**
3. Search for SuperStore
4. Select the **SuperStore**



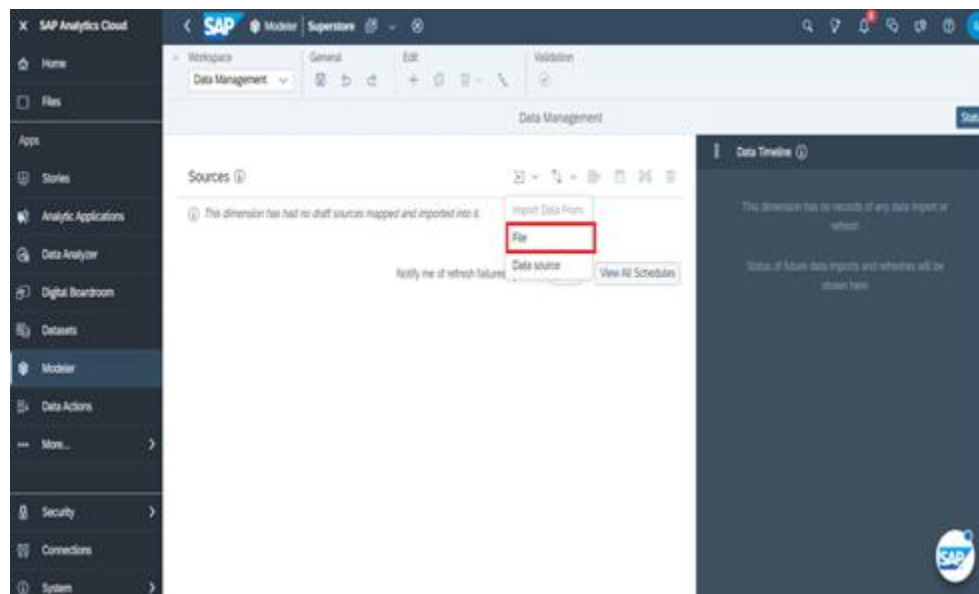
Step-2: Import the data

1. Click on the Down Arrow
2. Click on the Data Management
3. Click on the **import** option



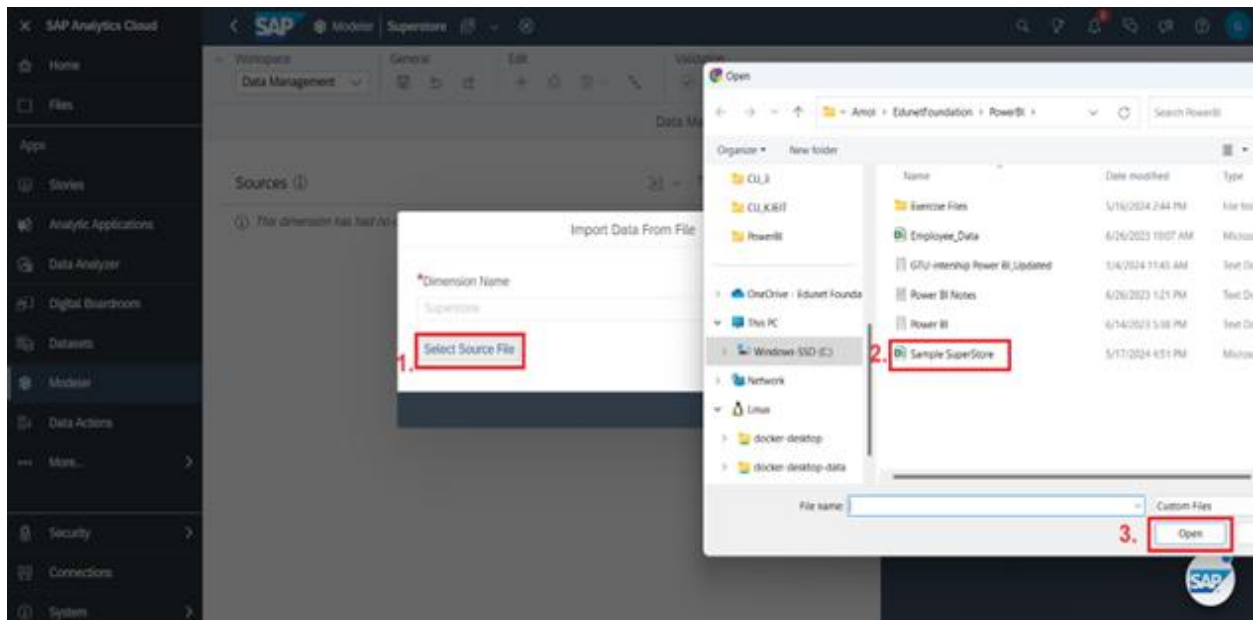
Step-2: Import the data

1. Click on **File**



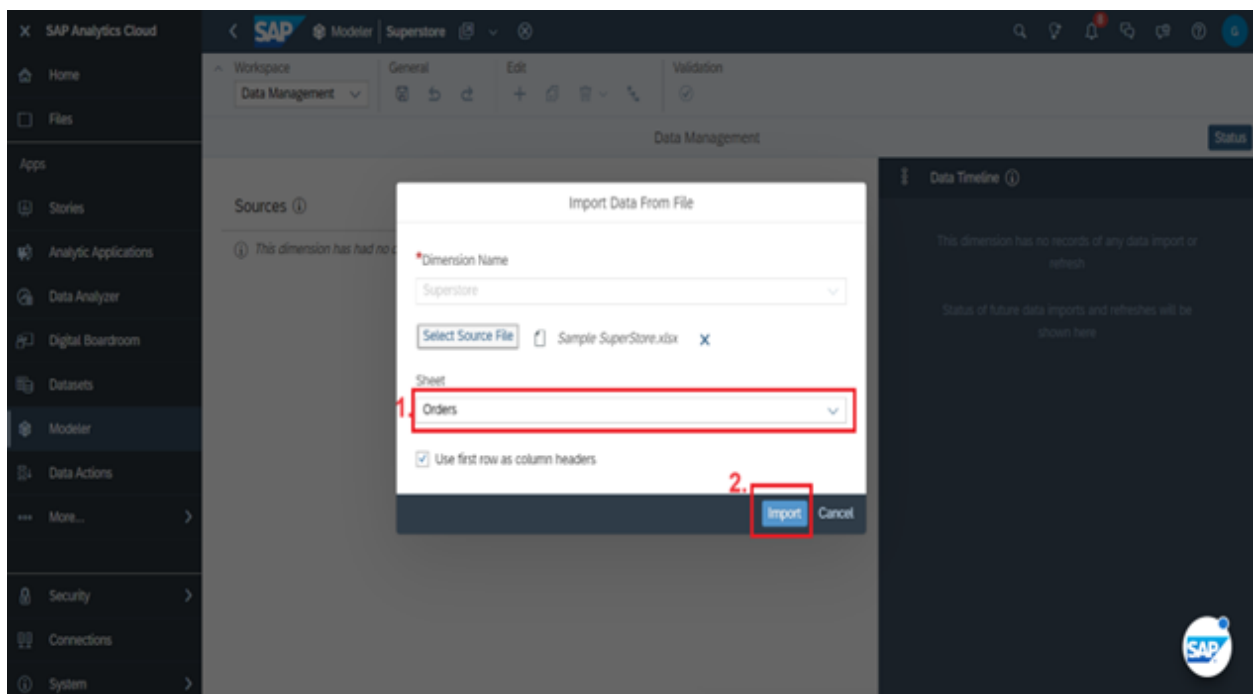
Step-2: Import the data

1. Select the **Source File**
2. Select the **Sample SuperStore** dataset
3. Click on the **Open**



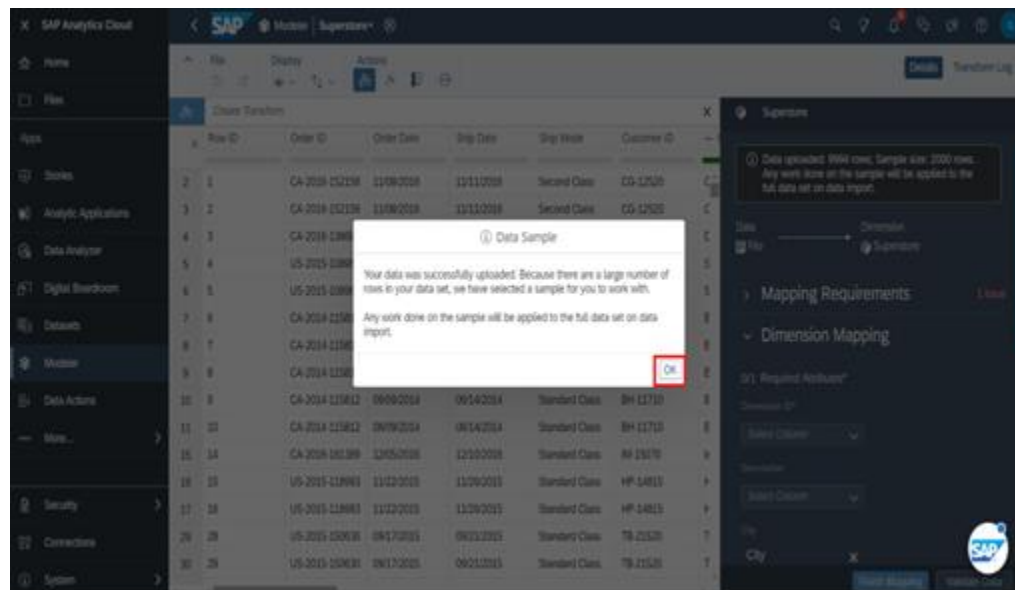
Step-2: Import the data

1. Select the **Orders**
2. Click on the **Import**



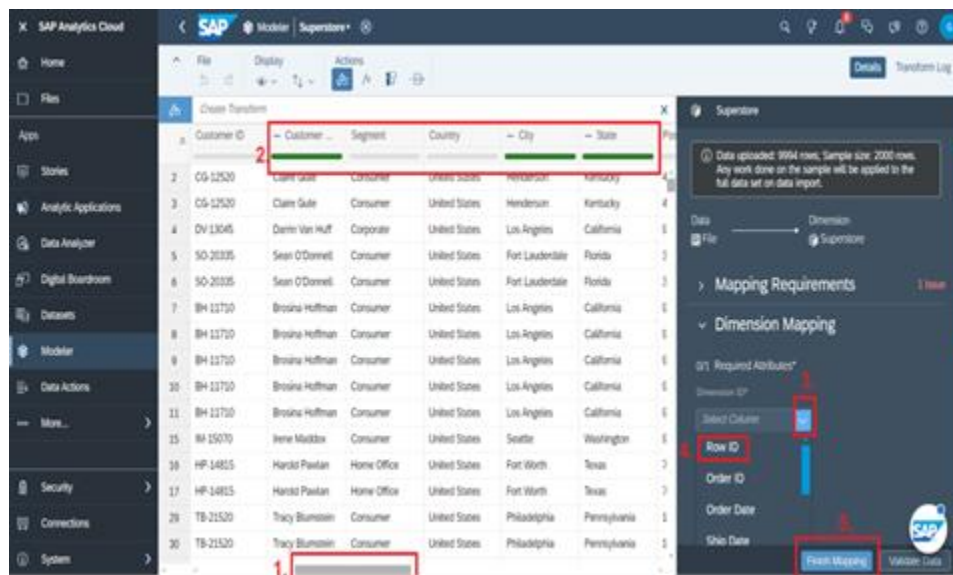
Step-2: Import the data

1. Click on the **OK** button



Step-2: Import the data

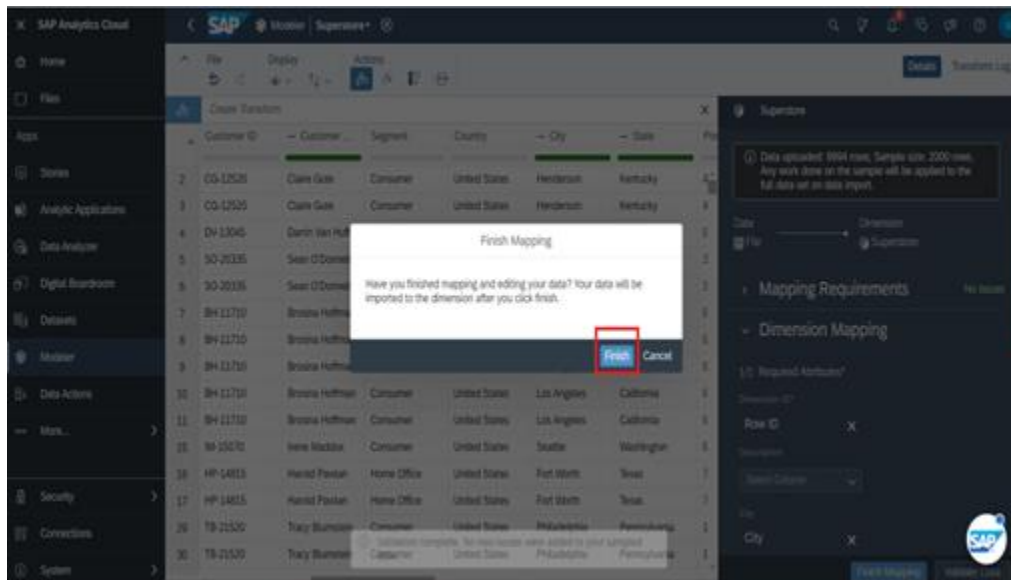
1. Scroll **horizontally** to right side
2. Check the columns in **green colour** indicate that column create in last practical's such as **Customer Name, City, State and the Sales**
3. Click on the **Down Arrow**
4. Click on the **Row ID**. Here this column values should be unique
5. Click on the **Finish Mapping**



Step-2: Import the data

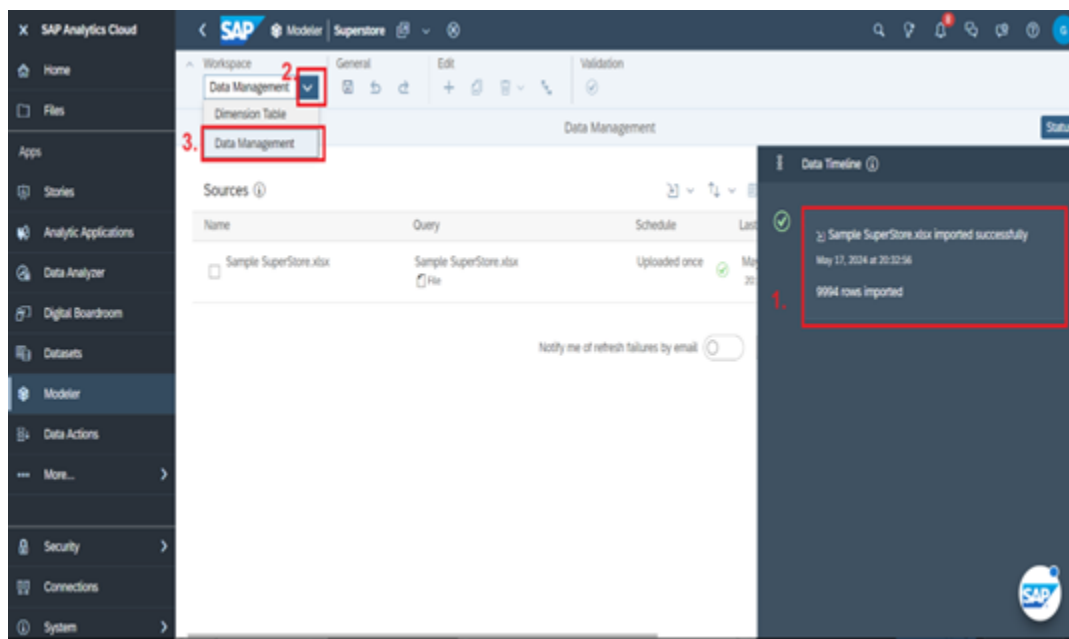
1. Click on **Finish** button

Disclaimer: The content is curated from online/offline resources and used for educational purpose only



Step-2: Import the data

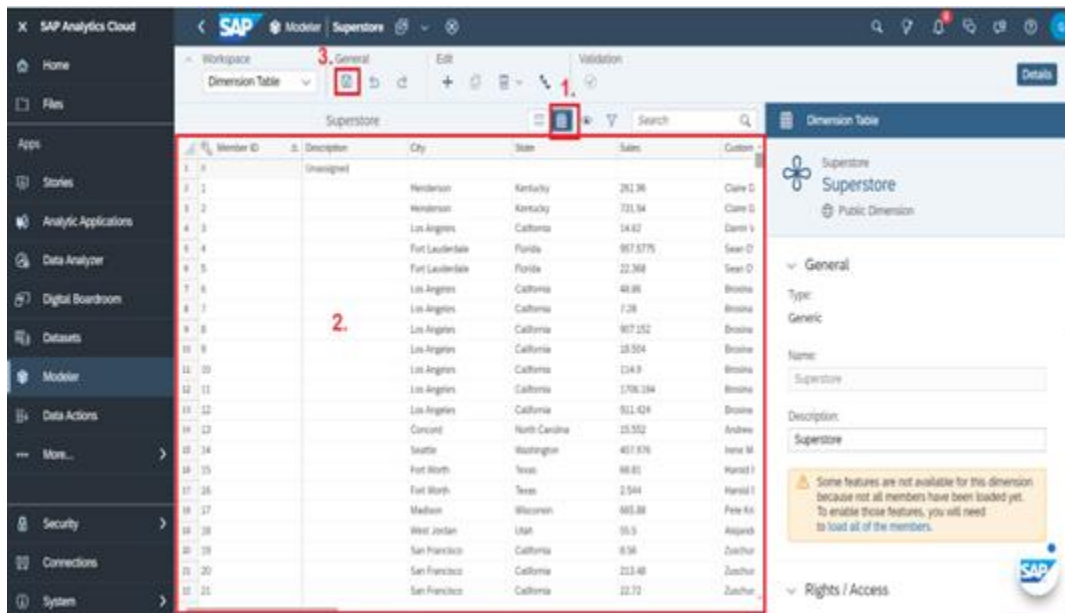
1. See the data imported successfully
2. Click on the **Down Arrow**
3. Click on the **Data Management**



Step-3: View the data

1. Click on the **Table View** “switch display mode to grid”
2. You can view the all the data imported successfully

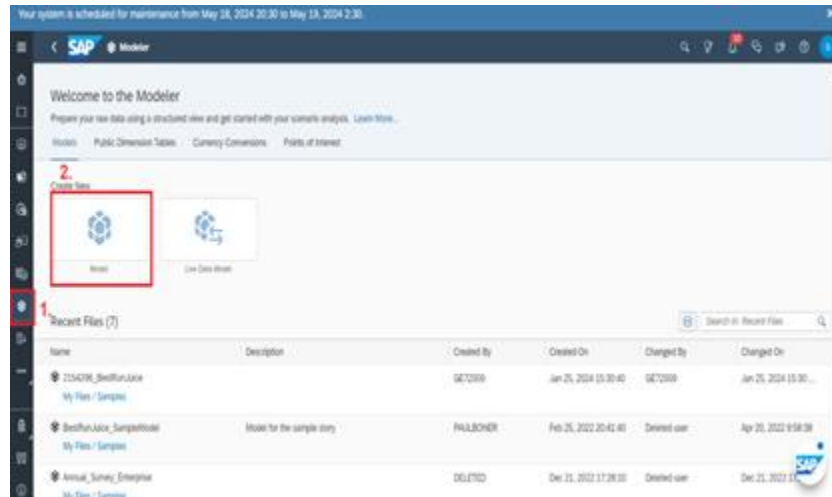
- Click on the **Save** button



6.4 Practical - Import Create an Import Analytic Model

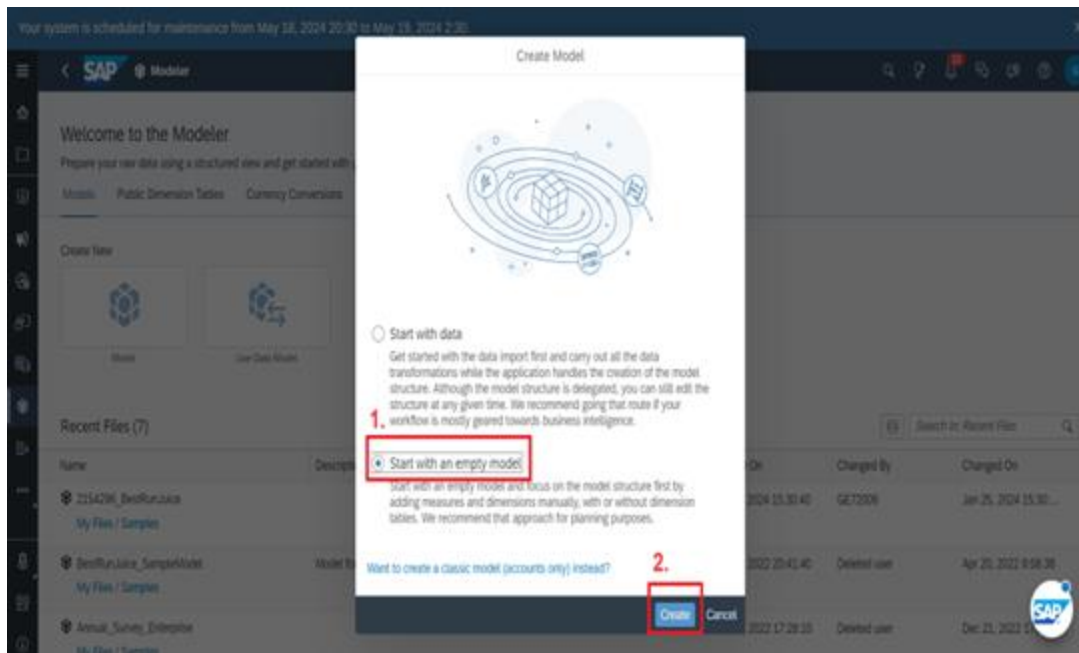
Step-1: Create Model

- Select the **Modeler**
- Click on the **Model**



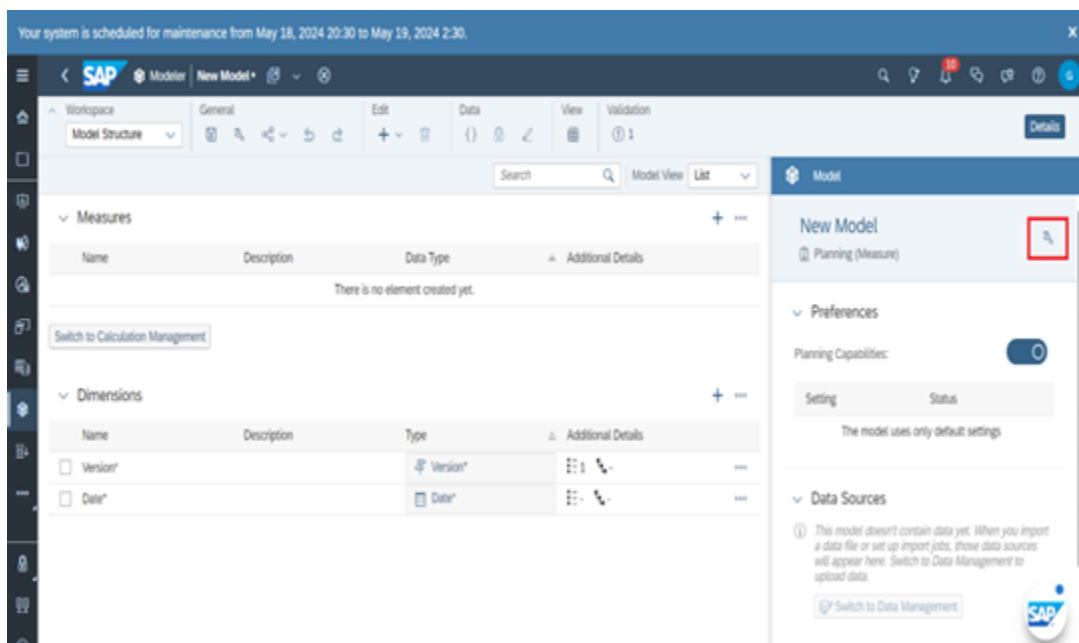
Step-1: Create Model

- Select the **Start with an empty model**
- Click on the **Create**



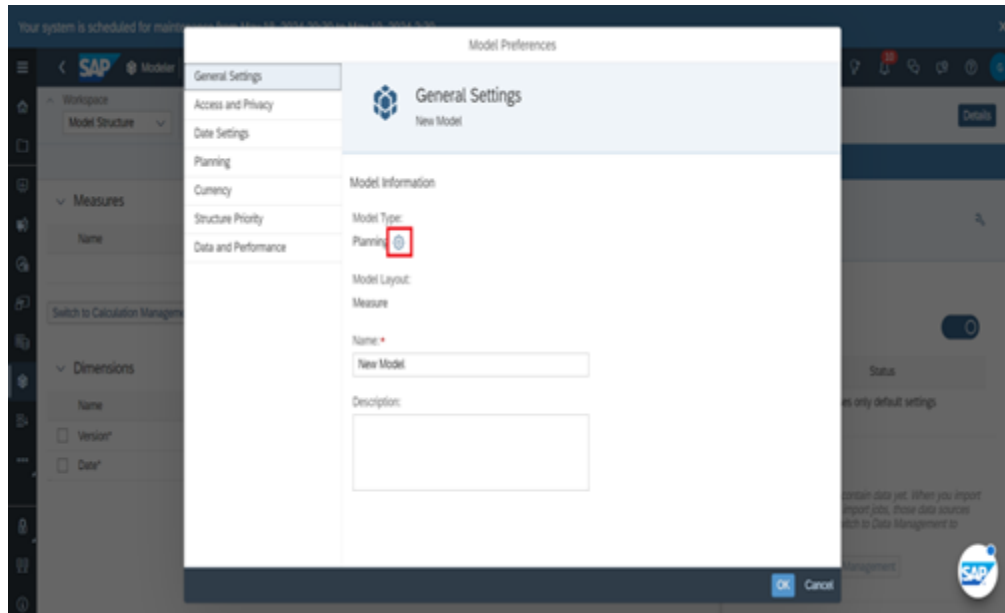
Step-1: Create Model

1. Select the **Model Preferences**



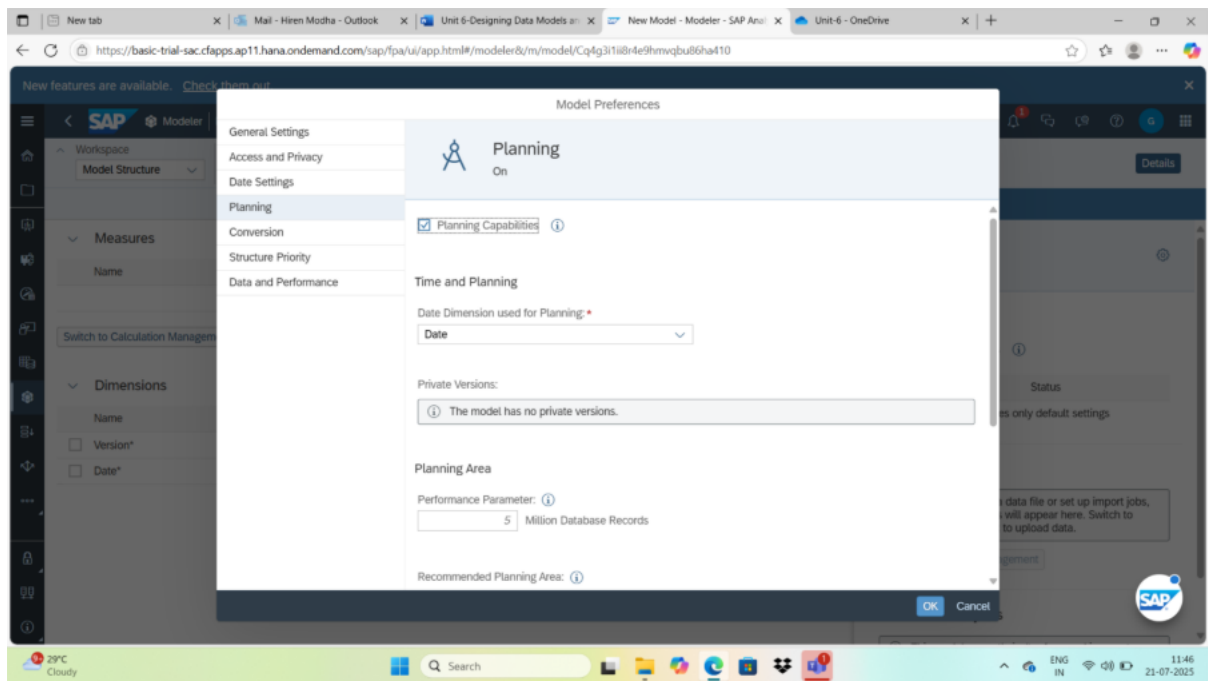
Step-1: Create Model

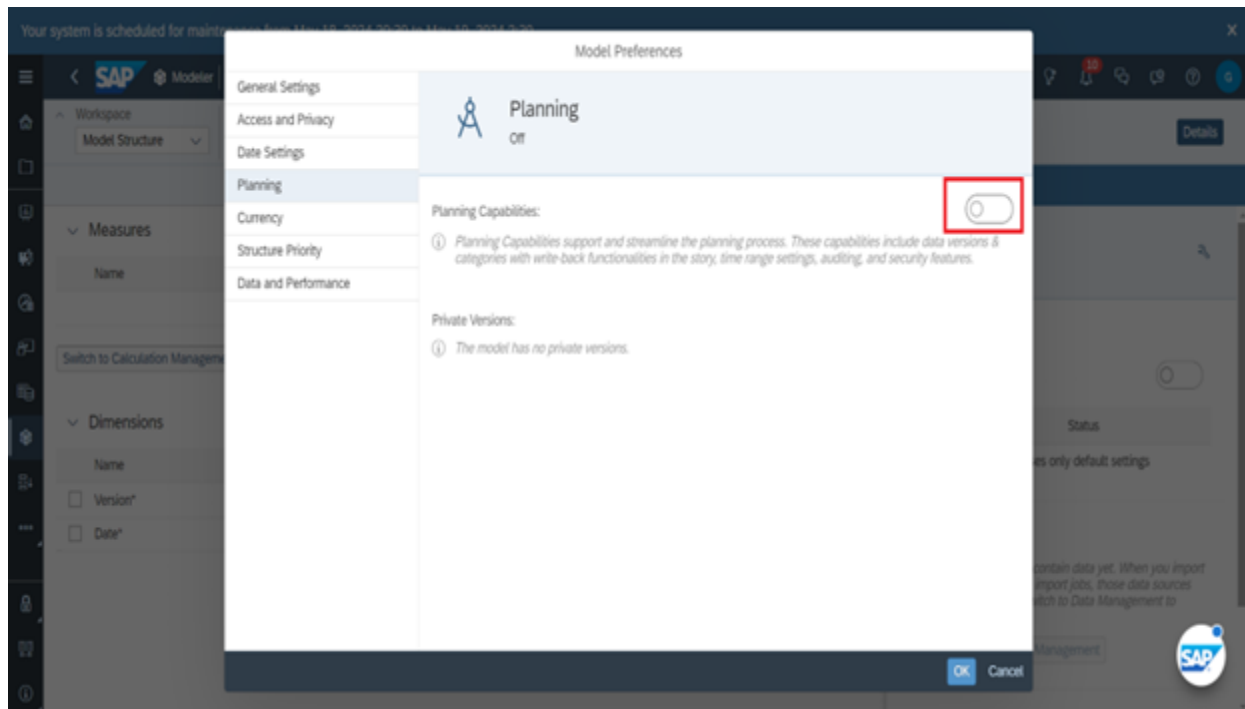
1. Click on the **Settings**



Step-1: Create Model

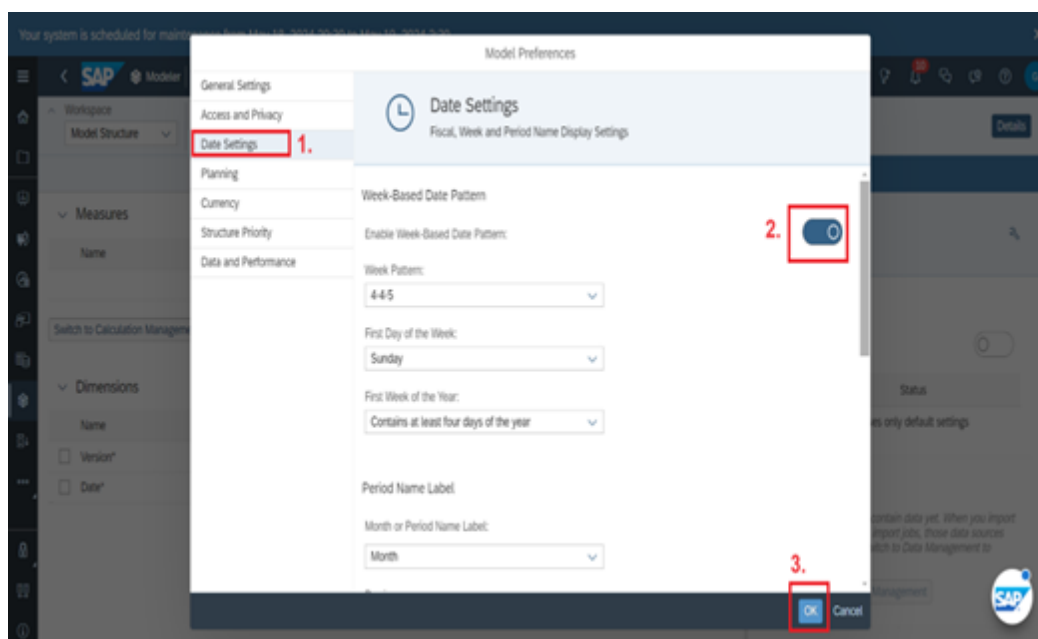
1. Click on the **Switch Off Planning Capabilities**





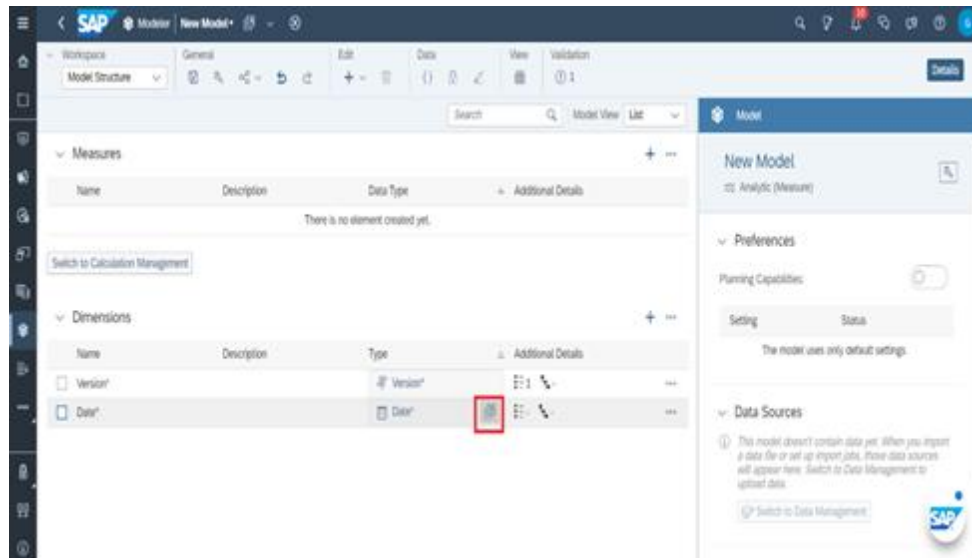
Step-1: Create Model

1. Click on the **Date Settings**
2. Enable the Week Based Data Pattern



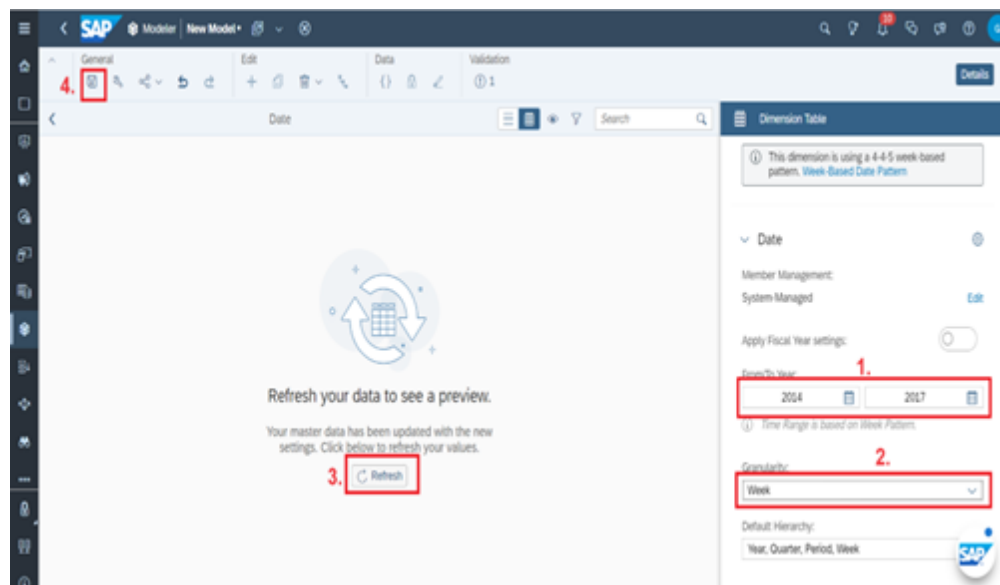
Step-1: Create Model

1. Click on  the **Edit Dimension Table**



Step-1: Create Model

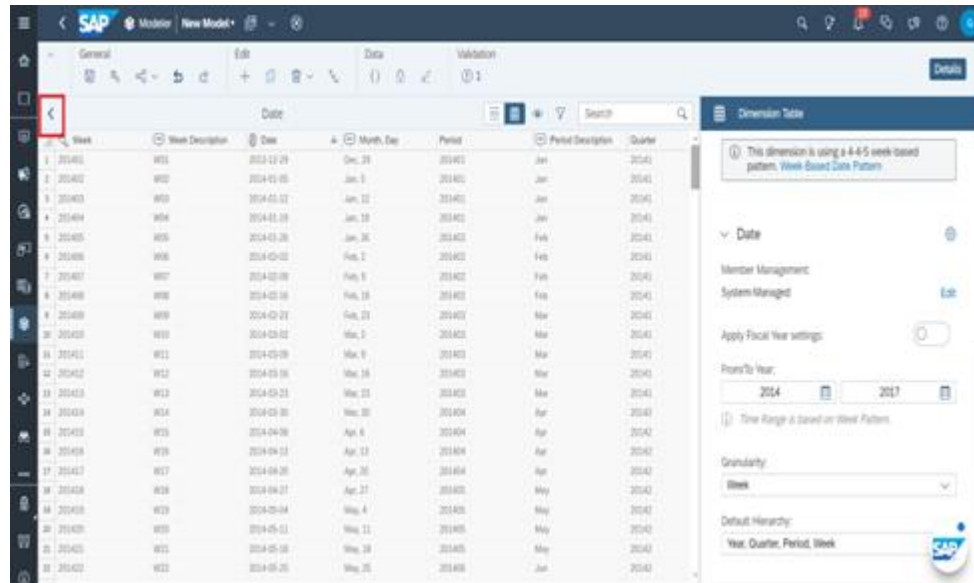
1. Select the calendar icon and select **2014** and **2017** year
2. Select the **Week** in Granularity
3. Click on the **Refresh**



Step-1: Create Model

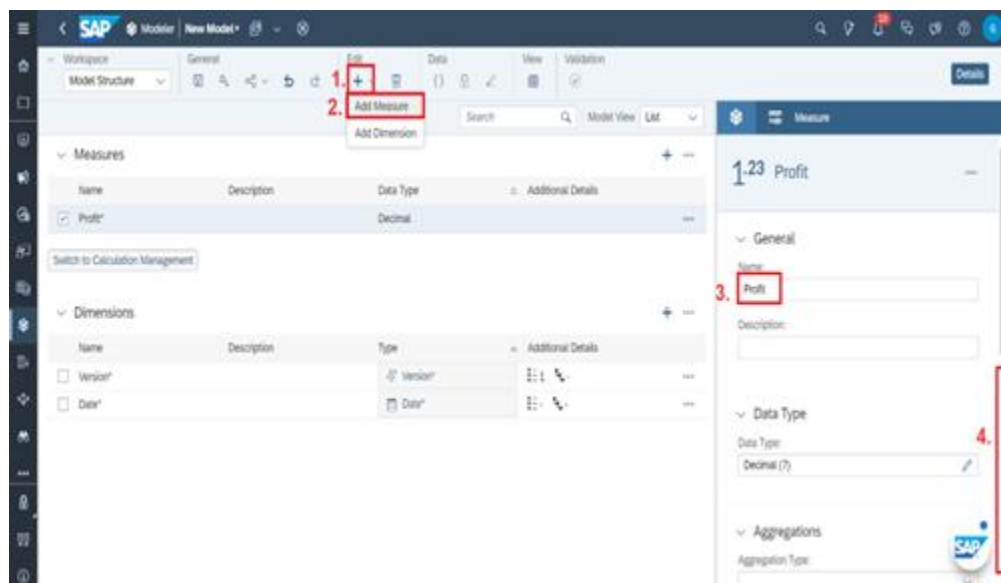
You can see the date column is spitted into different format

1. Click on the **back** option



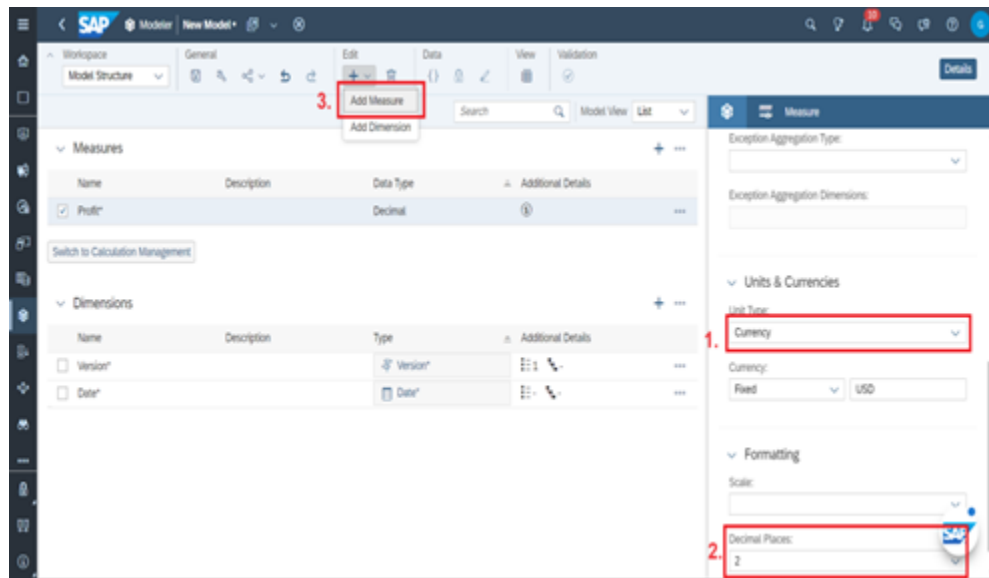
Step-2: Create Measure

1. Click on the **+**
2. Click on the **Add Measure**
3. Enter the **Profit** in Name field
4. Scroll down



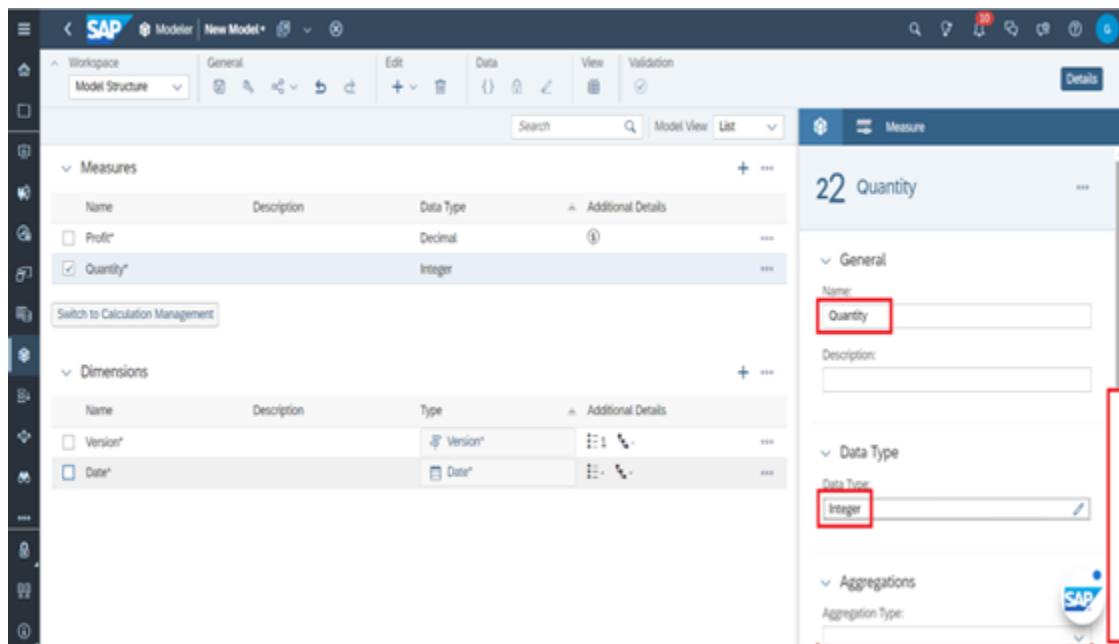
Step-2: Create Measure

1. Select the Unit type as **Currency**
2. Select the **2** in Decimal places
3. Click on the **+** and Click on the **Add Measure**



Step-2: Create Measure

1. Enter the **Quantity** in the Name
2. Select the **Integer** in Data types
3. Scroll down



Step-2: Create Measure

1. Select the **Unit** in the Unit type
2. Create new label **PC** in the Fixed Unit

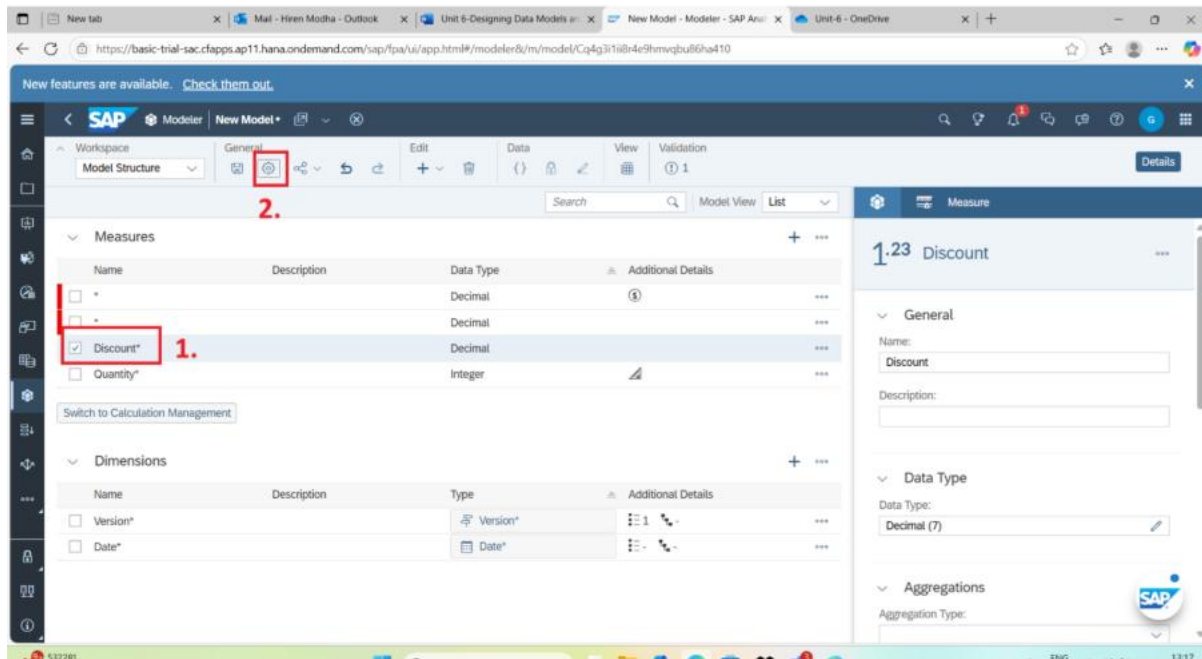
3. Similarly Create new measure as **Discount**

The top screenshot shows the SAP Modeler interface with the 'Measures' table. The 'Quantity' measure is selected. A red box highlights the '+' icon in the toolbar, labeled '3.'. The 'Measure' configuration panel on the right shows the 'Unit' dropdown highlighted with a red box and labeled '1.', and the 'Fixed Unit' dropdown highlighted with a red box and labeled '2.'.

The bottom screenshot shows the same interface but with the 'Measure' configuration panel expanded. The 'Unit' dropdown is highlighted with a red box and labeled '1.', and the 'Fixed Unit' dropdown is highlighted with a red box and labeled '2.'.

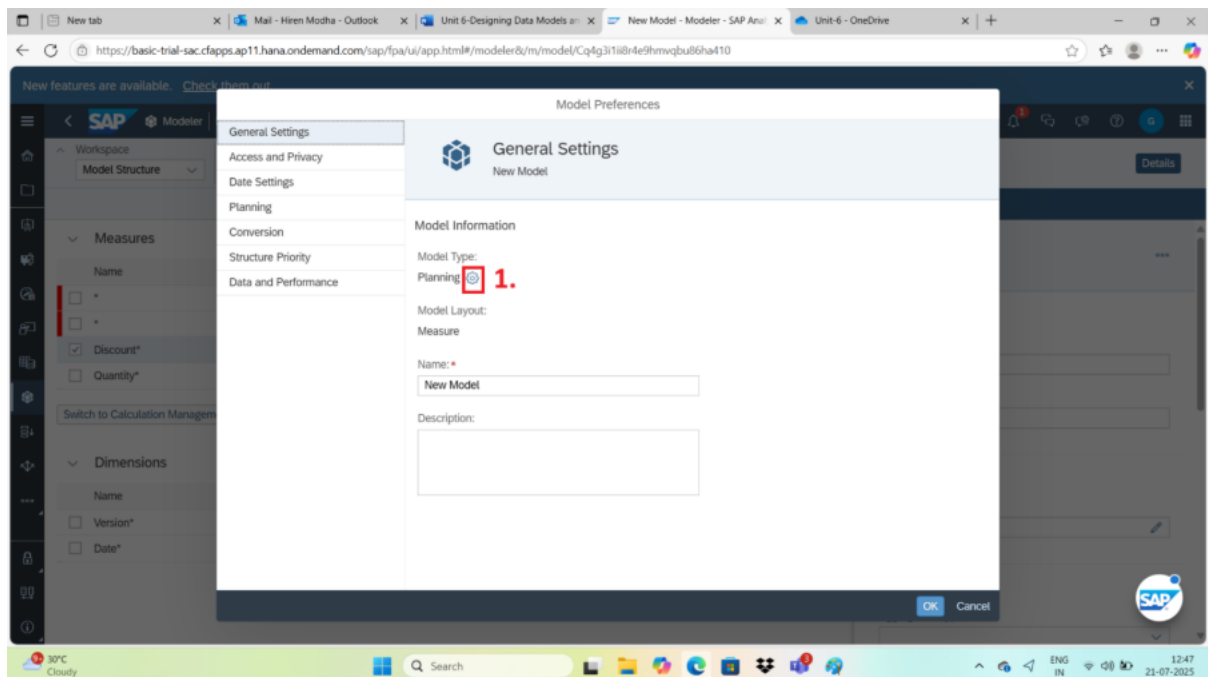
Step-2: Create Measure

1. See **Discount** measure created
2. Click on the **Model Preference**

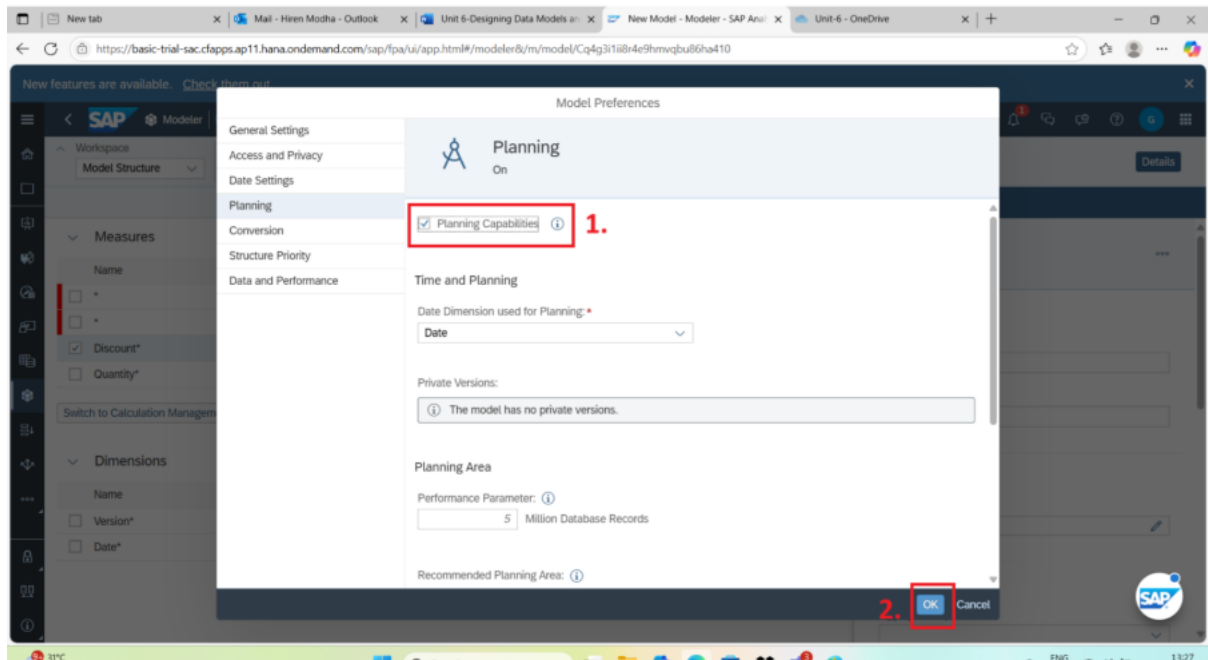


Step-2: Create Measure

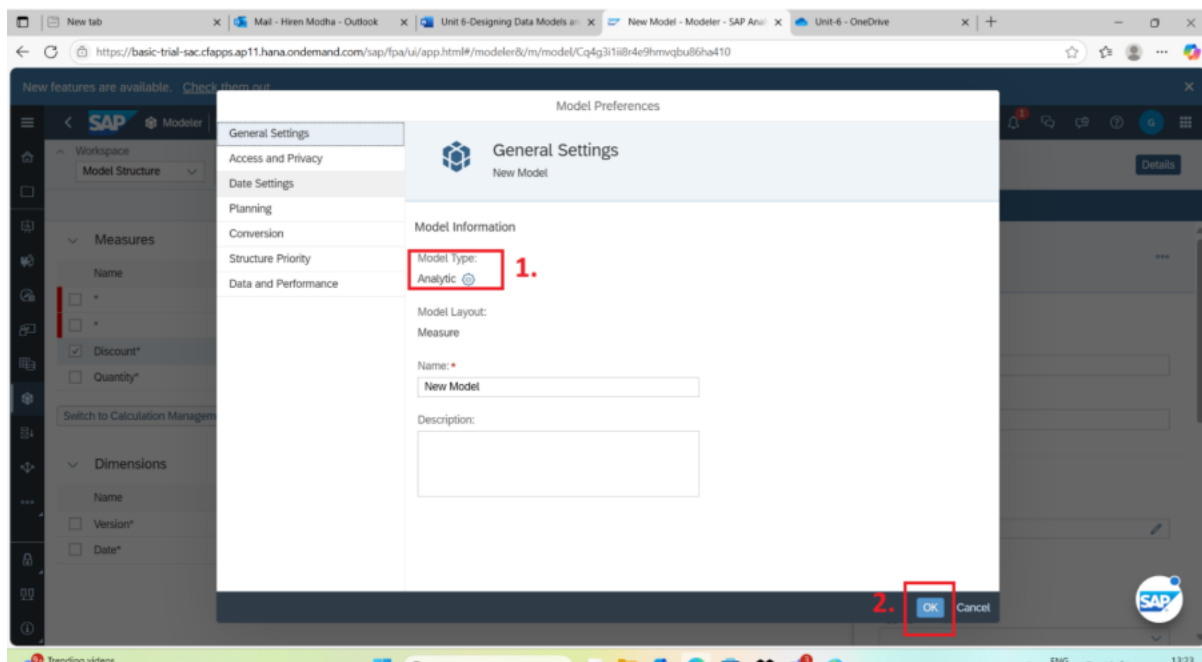
1. See **Model Type** is planning.
2. Go to setting.



3. Uncheck Planning Capabilities to set to **Analytic** then press OK.

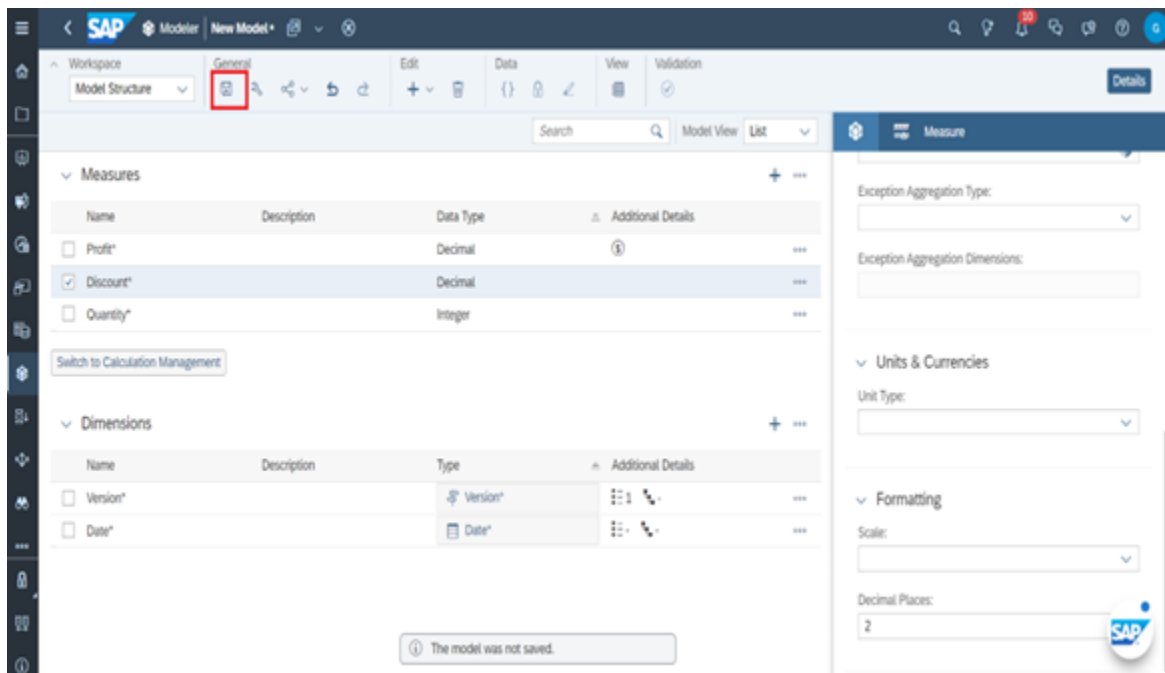


4. Again, click on **Model Preference** as shown in step 2, ypu will see Model type is **Analytic**.



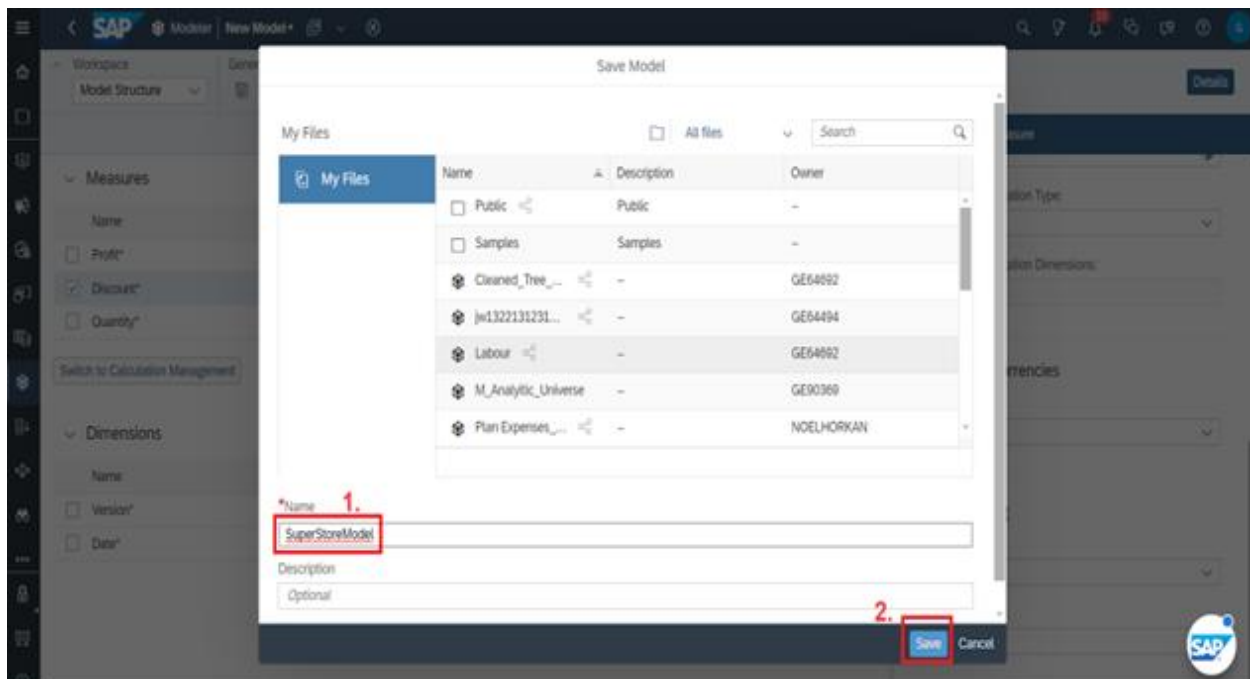
Step-3: Save the model

Click on **Save**



Step-3: Save the model

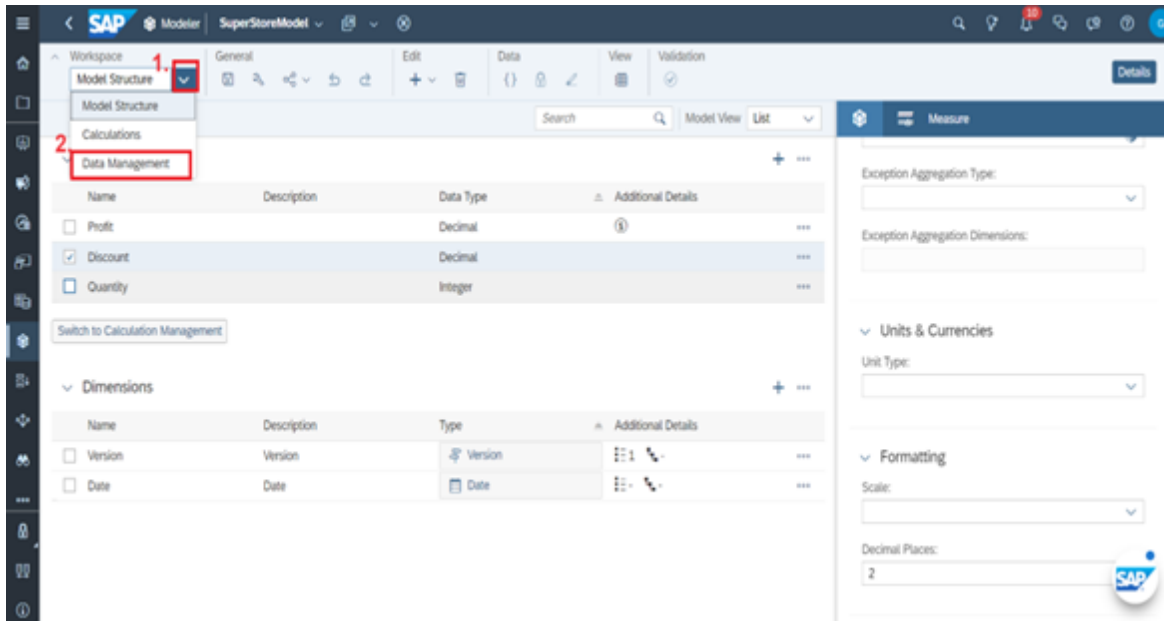
1. Enter the Name as **SuperStoreModel**
2. Click on **Save**



6.5 Practical - Import and Prepare Data in an Analytic Model

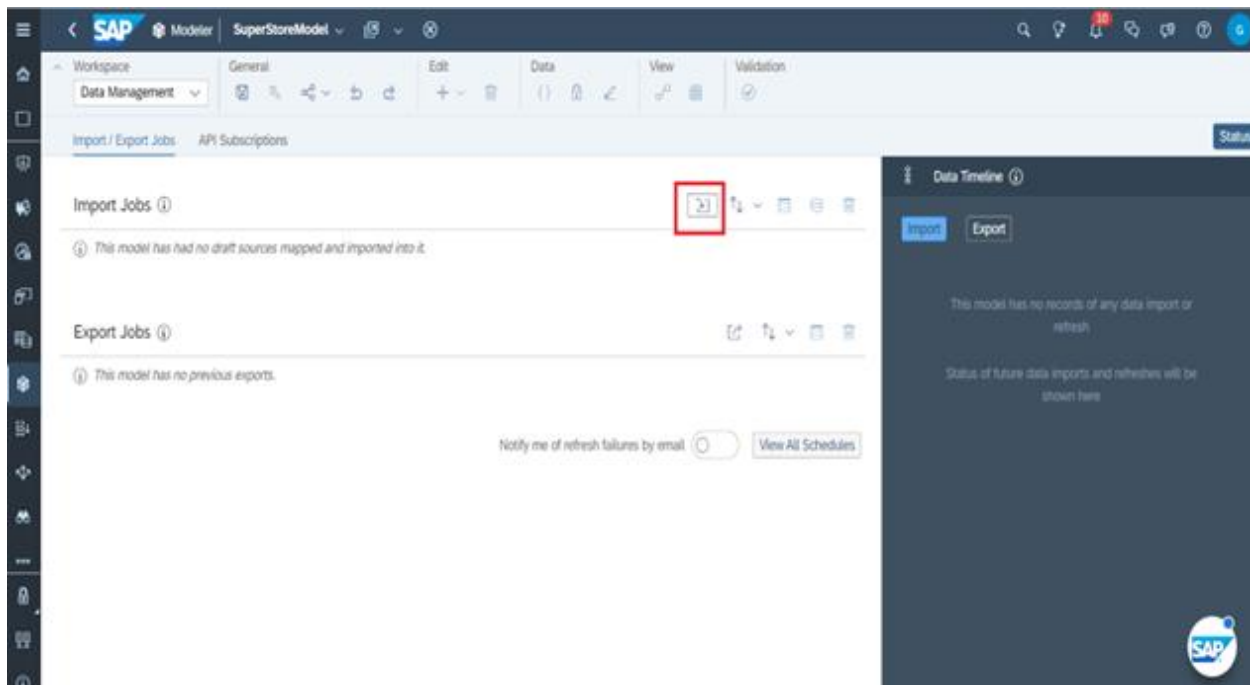
Step-1: Import Data

1. Click on down arrow
2. Select the **Data Management**



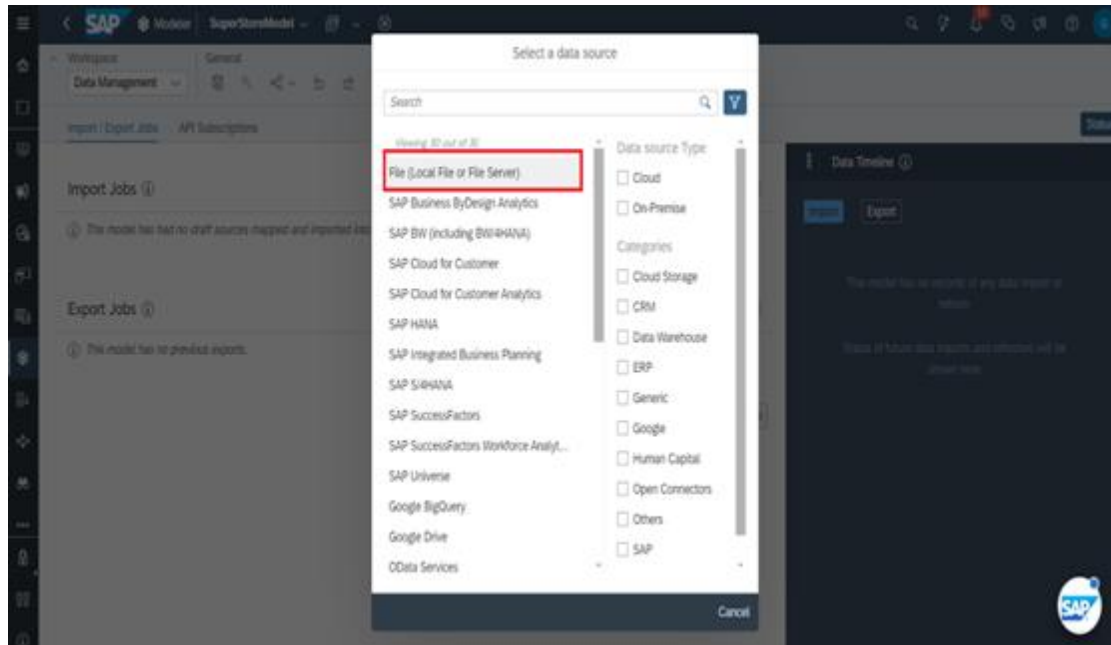
Step-1: Import Data

1. Click on **Import Data**



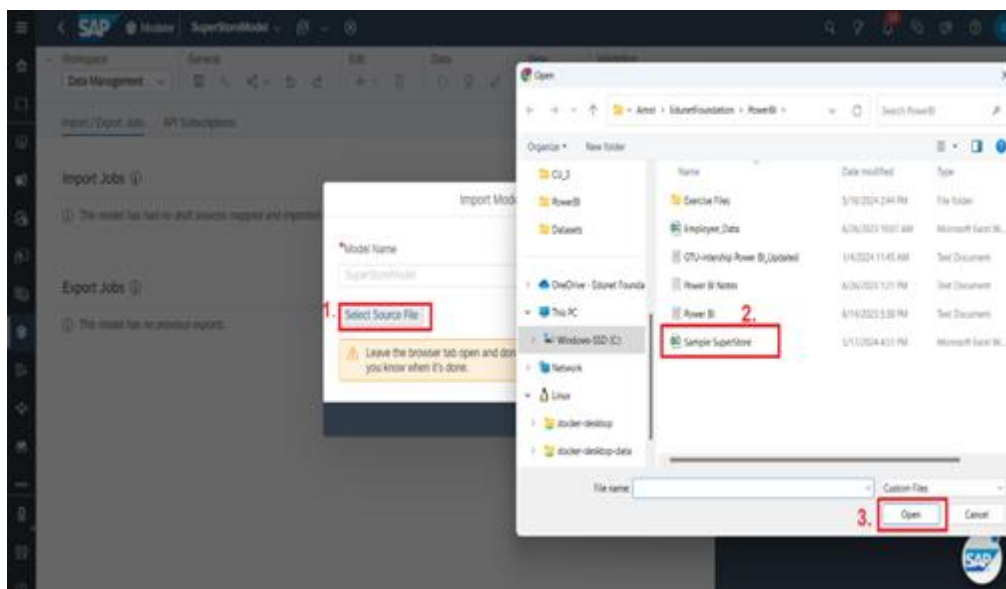
Step-1: Import Data

1. Click on **File (Local File or File server)**



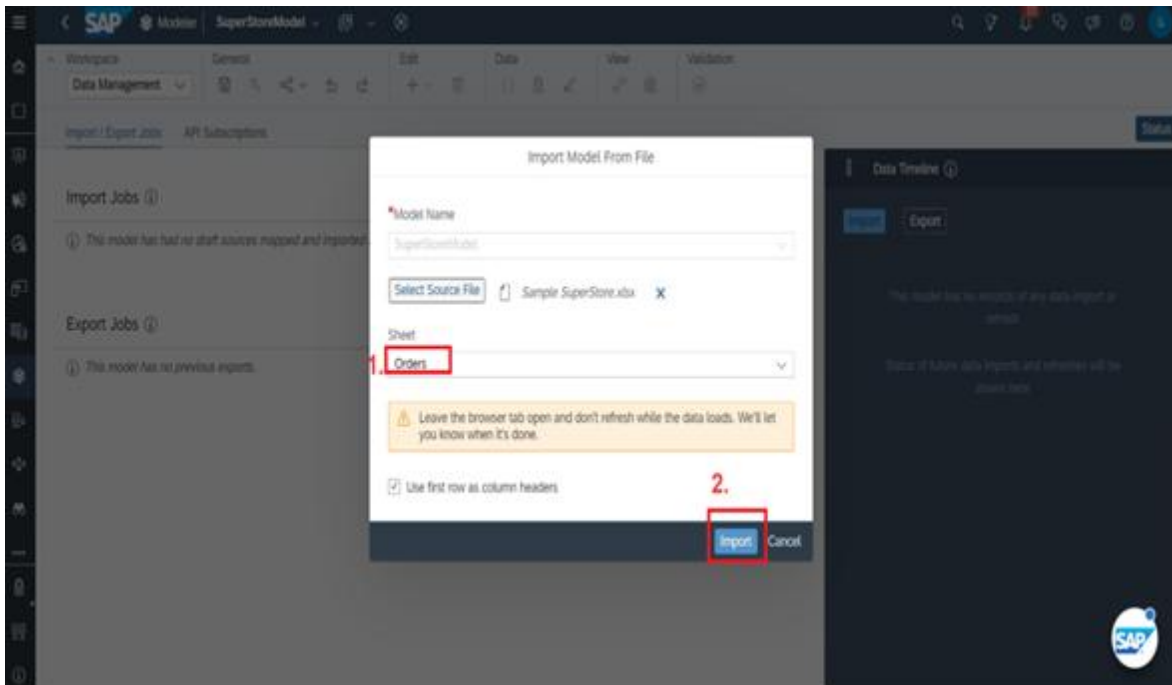
Step-1: Import Data

1. Click on the **Select Source File**
2. Select the Sample **SuperStore** file
3. Click on **Open**



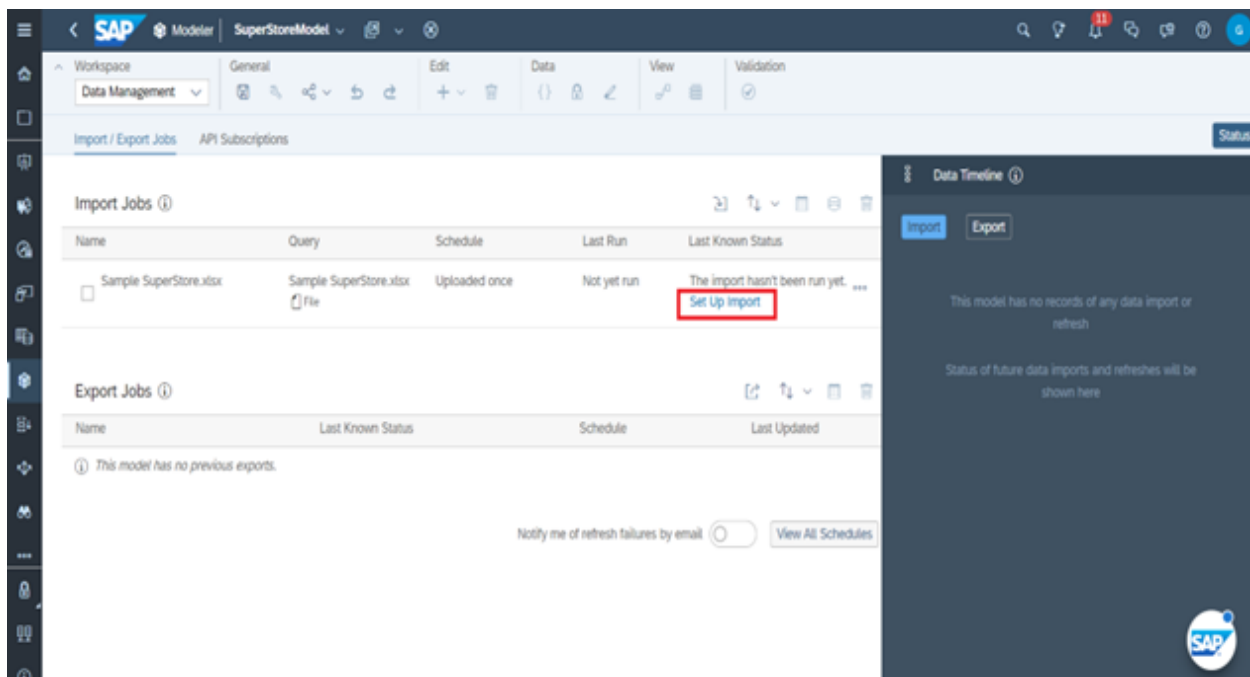
Step-1: Import Data

1. Select the **Orders**
2. Click on **Import** button



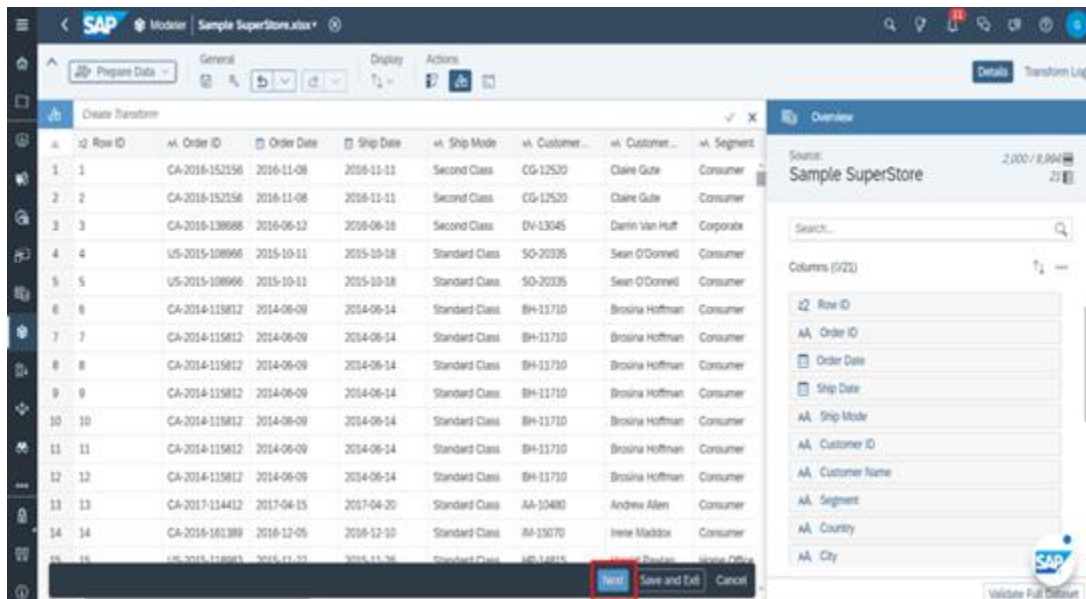
Step-1: Import Data

1. Click on **Set up the import**



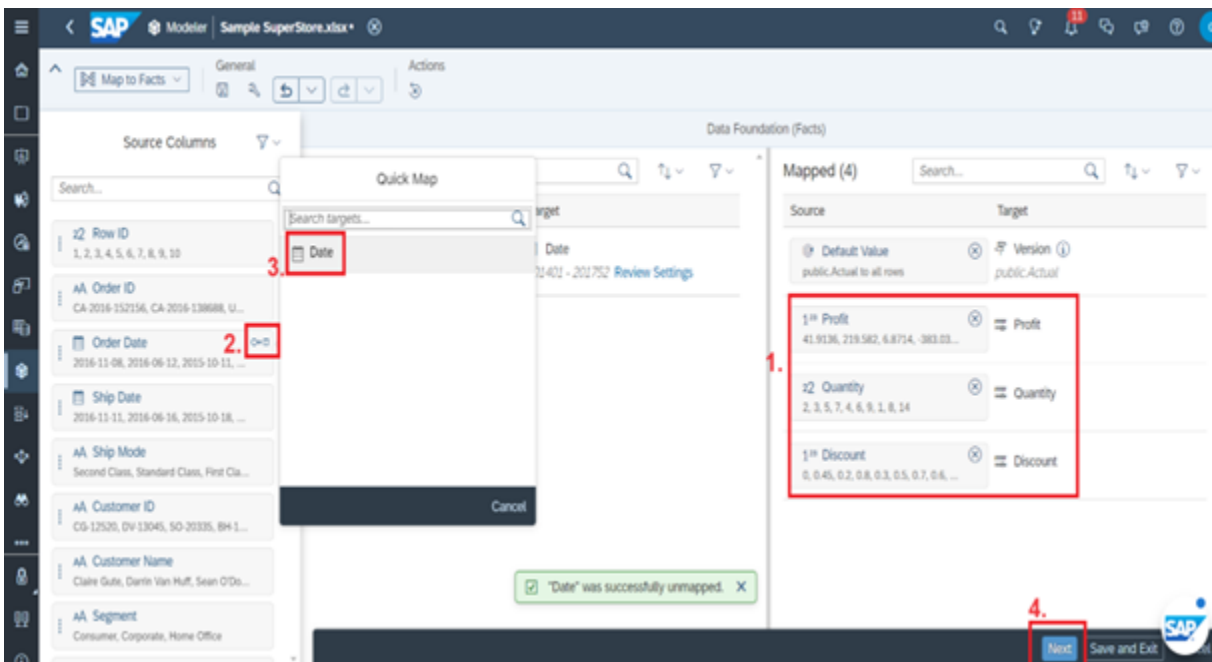
Step-1: Import Data

1. Click on **Next** Button



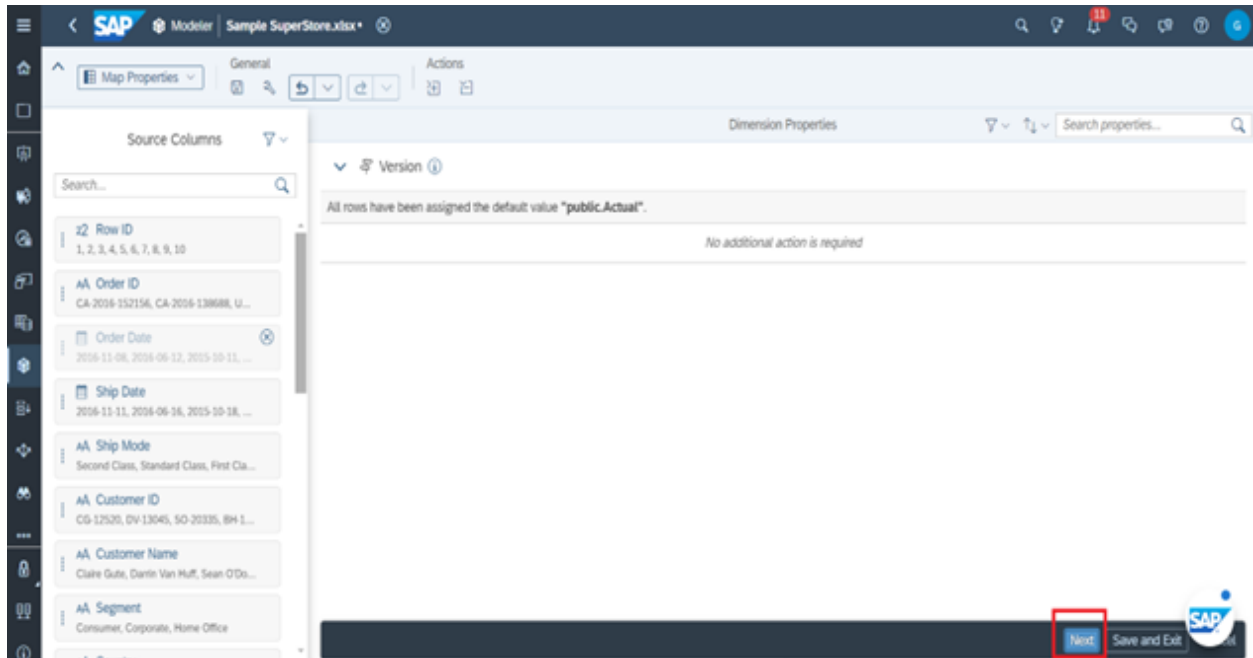
Step-2: Map to Facts

1. See all the data automatically map
2. Click on **Quick Map Order Date**
3. Select **Date**
4. Click on the **Next** button



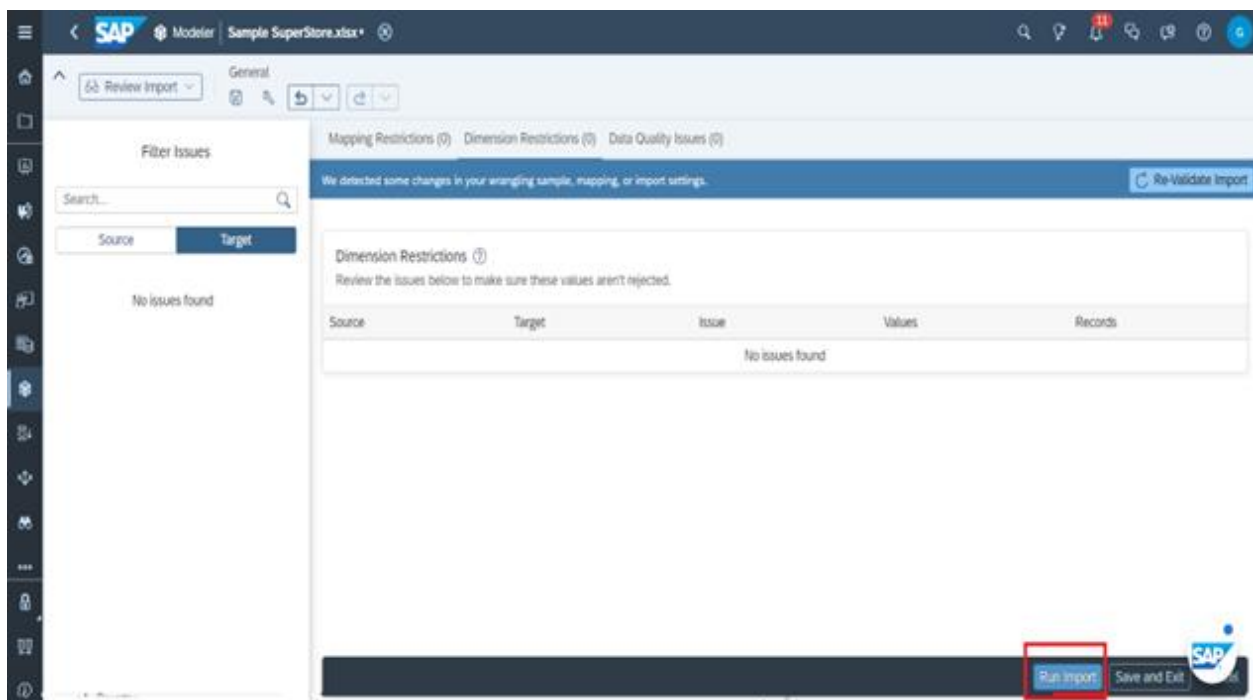
Step-2: Map to Facts

1. Click on the **Next** button



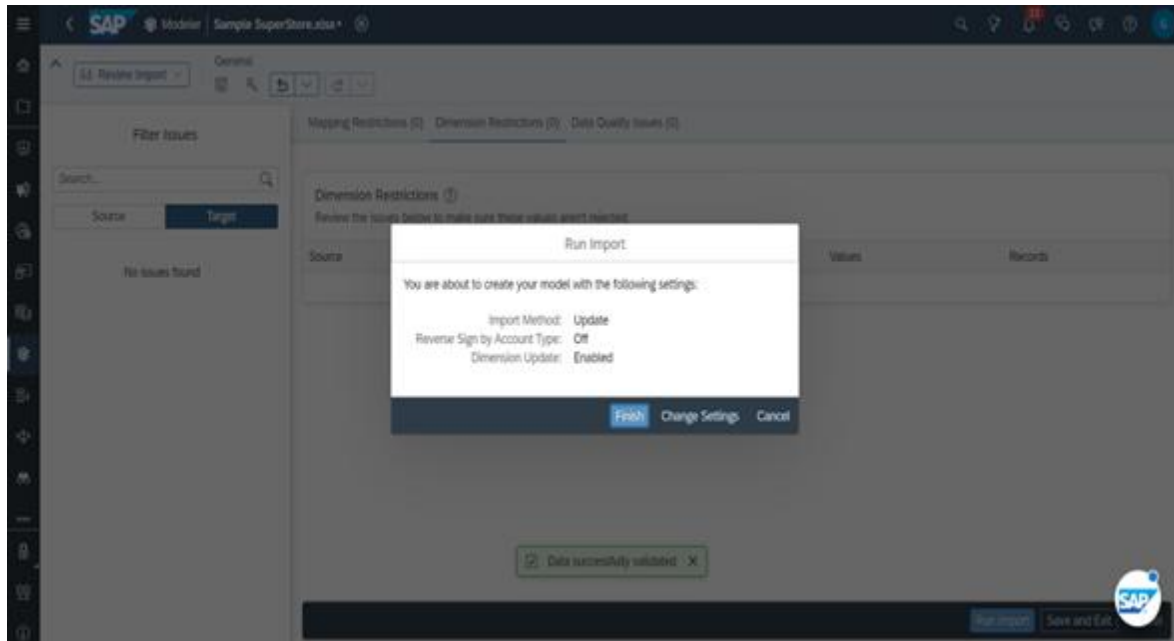
Step-2: Map to Facts

1. Click on the **Run Import** button
2. You will get warning so click on the **Validate the data**



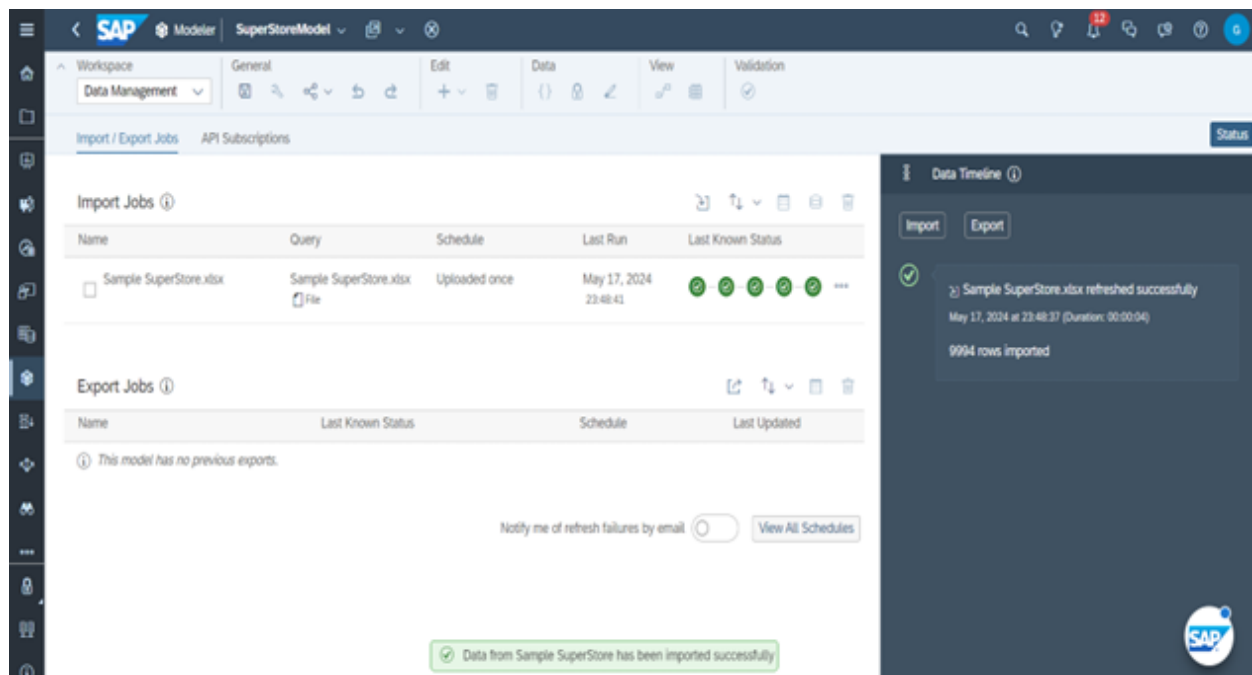
Step-2: Map to Facts

1. Click on the **Finish**



Step-2: Map to Facts

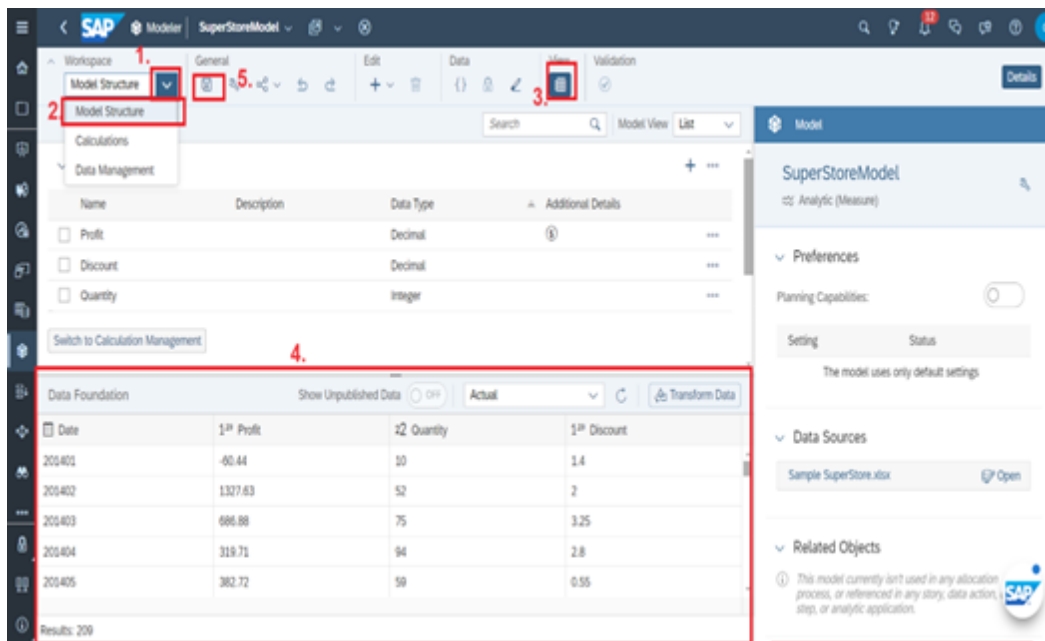
You can see Successfully imported the data



Step-3: View the data

1. Click on the **down arrow**
2. Click on the **Model Structure**
3. Click on the **Data Foundation view**
4. You will get the data
5. Click on the **Save** button

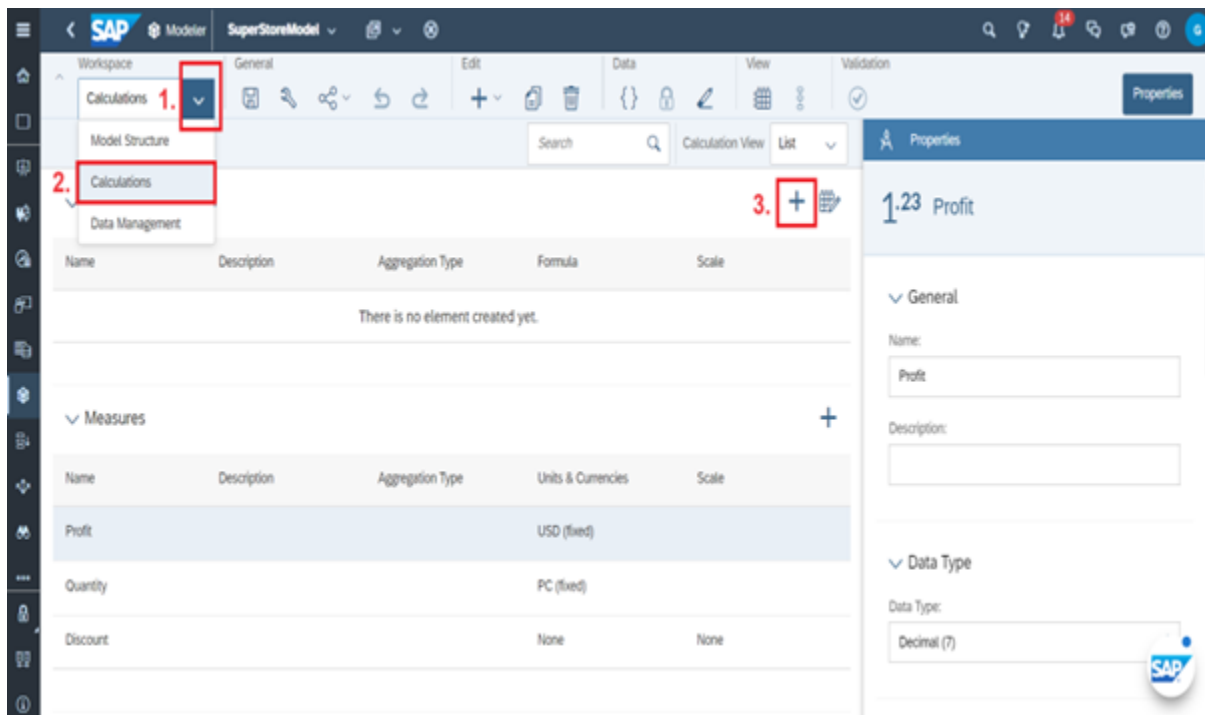
All the data map in the weeks see it's showing Result as **209**.



6.6 Practical - Create a Calculated Measure in a Model

Step-1: Calculated Measure

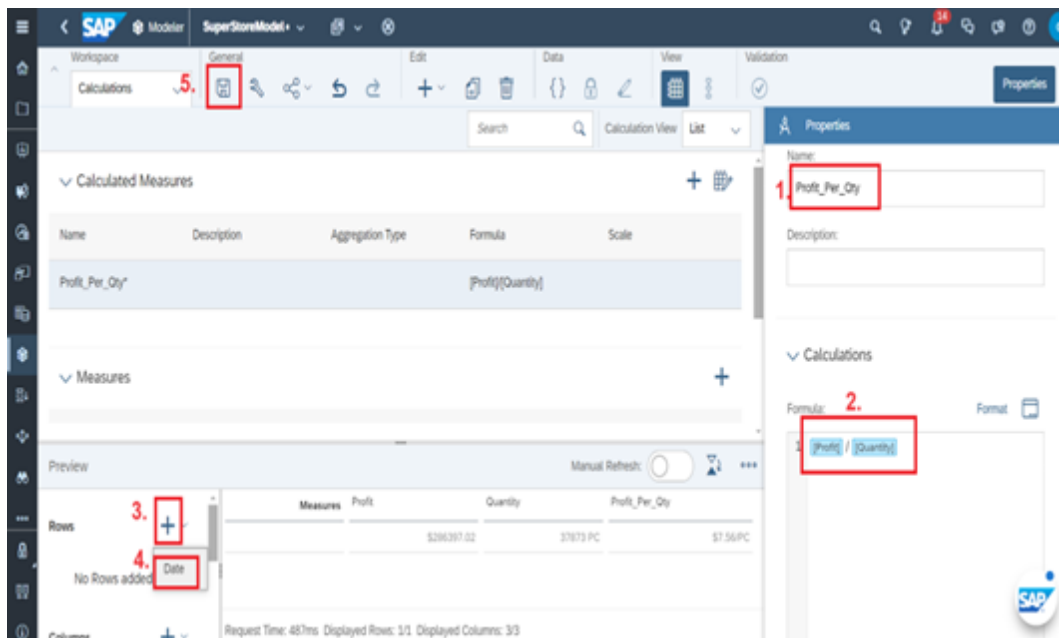
1. Click on the **down arrow**
2. Click on the **Calculations**
3. Click on the **+** You will get the data



Step-1: Calculated Measure

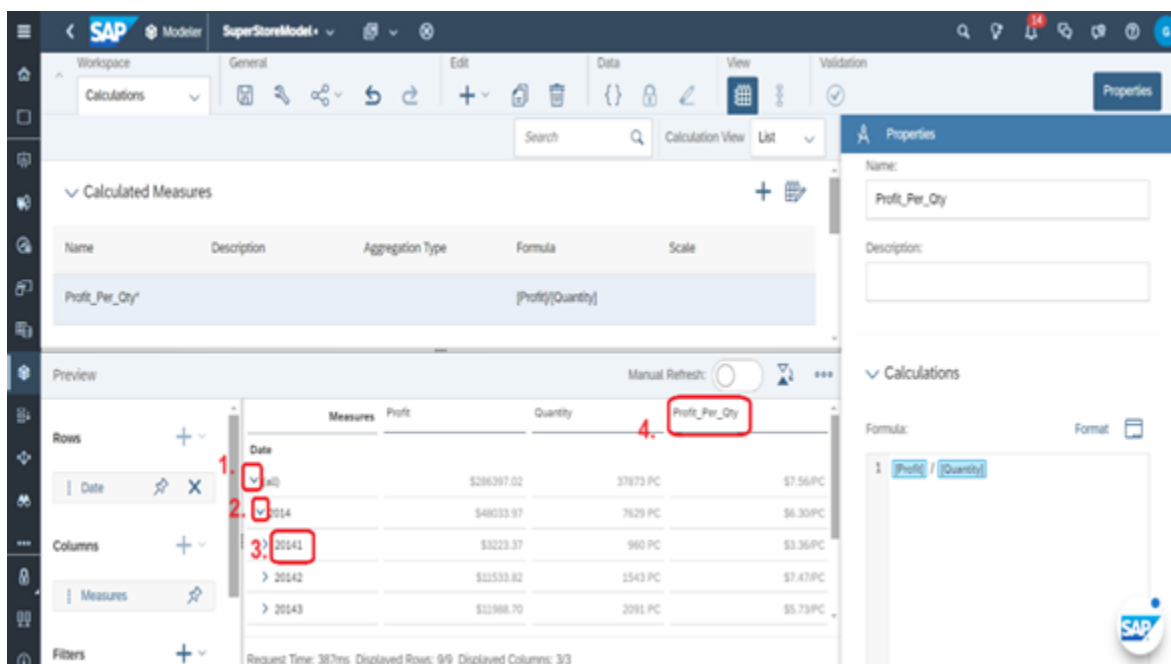
1. Enter **Profit_Per_Qty** under the Name
2. Type **profit/Quantity** under the formula
3. Click on the +
4. Click on the **Date**
5. Click on the **Save**

Then the table will display



Step-1: Calculated Measure

1. Click on **Down Arrow**
2. Click on **Down Arrow**
3. See we will get the year and 1st week data
4. Check **Profit_Per_Qty** calculated measure added



R